

Where do Gradient Methods Really Happen? Maybe Not Where We Expected

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Abstract—We study the paper Song et al. (2025), which examines the hypothesis that neural network training takes place in low-dimensional subspaces. The authors show that the observed alignment of the gradient with a dominant subspace does not necessarily imply that the principal learning dynamics occur within that subspace.

Within the scope of this project, we first plan to reproduce the main results of the paper for the toy model, MLP, CNN, and Transformer settings. We then intend to extend the study by considering alternative optimizers, other low-dimensional subspaces, and the effect of architectural properties of neural networks on the observed phenomenon.

I. INTRODUCTION

In modern deep learning, model parameter spaces are high-dimensional; however, recent studies show that effects associated with low-dimensional spaces are observed during training (Gur-Ari et al., 2018; Song et al., 2025; T. Li et al., 2021). In the works (Gur-Ari et al., 2018; Song et al., 2025) the authors study a phenomenon in which, in a k -class classification problem, the gradient aligns with the dominant subspace after several training steps. The hypothesis that all learning takes place in this subspace was refuted in Song et al. (2025). The problem of identifying a low-dimensional subspace in which training can be performed is also considered in other lines of work, which are discussed in more detail in the literature review.

The goal of this work is to examine the gradient alignment phenomenon across a broader range of optimizers, such as Muon, forward-gradient methods, among others, as well as across different low-dimensional subspaces. In addition, we aim to provide a theoretical justification for some of the empirically obtained results.

II. RELATED WORK

The scientific landscape surrounding this topic naturally falls into several lines of work.

The first and most closely related line consists of works on the Hessian spectrum and on the alignment of the gradient with its top eigenspace. Early studies (Sagun, Bottou, et al., 2017; Sagun, Evci, et al., 2018; Ghorbani et al., 2019), already showed that the Hessian spectrum in deep networks typically has a pronounced structure: a bulk concentrated near zero

and a small number of outlier eigenvalues. Against this background, the work Gur-Ari et al. (2018) formulated a strong hypothesis: after a short initial phase, the gradient aligns with a small subspace spanned by several top Hessian eigenvectors. A more recent theoretical work Arous et al. (2024) showed, using simplified high-dimensional mixture models, that a similar alignment can arise in a mathematically rigorous way. However, it was precisely the paper Song et al. (2025) that provided a fundamental counterexample to a direct interpretation of this observation: when projecting the update onto the dominant Hessian subspace, training stops; in some cases, the loss even starts to increase, while the bulk component of the update remains surprisingly effective.

The latter paper provides the most important reference results for our project. First, it formalizes the dominant subspace as the span of the top- k Hessian eigenvectors, where the standard choice of k is the number of classes in the classification task. Second, it shows that for SGD in the gradient-flow regime, spurious alignment is caused primarily by stochastic noise, whereas for GD at the Edge of Stability, similar alignment is related to a self-stabilization mechanism. Third, in Appendix F, it demonstrates the same negative effect for SGDM and Adam. This led us to the next step: to determine to what extent this negative result depends on the choice of optimizer and on the way the subspace is defined.

The second line of literature concerns the more general idea that neural network problems have low effective dimensionality. The simplest formulation appears in C. Li et al. (2018): the network is trained not in the full parameter space, but in a fixed random subspace, and the minimal sufficient dimension is used as a measure of task complexity. In parallel, a stronger empirical hypothesis was proposed in T. Li et al. (2021), where the DLDR method was introduced and it was shown that some networks can be effectively trained in spaces with dimensions on the order of tens. These works clearly show that a negative result for one specific subspace does not close the topic of low-dimensional training as a whole.

Several related papers also belong to this line of work. In NLP Zhang et al. (2023) shows that fine-tuning pretrained language models can be interpreted through small task-specific subspaces. In few-shot learning, Gauch et al. (2022) proposes SubGD, an approach based on the eigendecomposition of the autocorrelation matrix of update directions. These works clearly indicate that, as an alternative to the Hessian top- k

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subspace, one can consider subspaces constructed not from curvature, but from the statistics of gradients or updates themselves.

III. PROBLEM STATEMENT

Let $\theta \in \mathbb{R}^p$ denote the vector of model parameters, and let

$$L(\theta) = \frac{1}{n} \sum_{i=1}^n l_i(\theta)$$

be the empirical loss function. At optimization step t , the current parameters are denoted by θ_t . Let g_t be either a stochastic or full gradient estimate, and let

$$u_t = \mathcal{O}_t(g_t, s_t)$$

be the raw update vector produced by an optimizer $\mathcal{O}$ with internal state s_t .

We consider a k -dimensional subspace $U_t \subset \mathbb{R}^p$ and denote by P_{U_t} the orthogonal projector onto this subspace. The update vector can then be decomposed into two orthogonal components:

$$u_t = P_{U_t} u_t + (I - P_{U_t}) u_t.$$

For a given optimizer $\mathcal{O}$ and a sequence of subspaces $\{U_t\}_{t \geq 0}$, we compare three optimization trajectories:

$$\theta_{t+1}^{\text{full}} = \theta_t - \eta_t u_t,$$

$$\theta_{t+1}^U = \theta_t - \eta_t P_{U_t} u_t,$$

$$\theta_{t+1}^{U^\perp} = \theta_t - \eta_t (I - P_{U_t}) u_t.$$

The goal is to analyze how different components of the optimizer update contribute to the optimization process. In particular, we study whether the part of the update lying in a selected low-dimensional subspace U_t is sufficient to reproduce the behavior of the full optimizer, and how much information is contained in the complementary component $U_t^\perp$.

This formulation is independent of the specific way in which the subspace U_t is constructed. Concrete choices of such subspaces are described separately below.

IV. SUBSPACE FAMILIES AND EXPERIMENTAL VARIANTS

In this work, we consider several families of subspaces U_t .

First, following the original dominant-subspace formulation, let

$$H_t = \nabla^2 L(\theta_t)$$

be the Hessian of the empirical loss at step t . We denote by $S_{t,k}^H$ the subspace spanned by the top- k eigenvectors of H_t , and by $P_{t,k}^H$ the corresponding orthogonal projector. This subspace captures the dominant curvature directions of the loss landscape.

Second, we consider a layerwise matrix representation of the momentum. For each layer, the corresponding momentum tensor is reshaped into a matrix $M_t^{(\ell)}$. We then compute its singular value decomposition and take the leading left and right singular vectors. The rank k is selected by the cumulative

energy of the singular values: we choose the smallest k such that

$$\frac{\sum_{i=1}^k \sigma_i^2}{\sum_i \sigma_i^2} \geq \varepsilon.$$

Equivalently, the energy threshold ε imposes an upper bound on the rank of the resulting approximation. This gives a low-dimensional dominant momentum subspace, which captures the main directions accumulated by the optimizer dynamics at the layer level.

For adaptive optimizers we use Adaptive Hessian projector and for Muon we use Muon-metric Hessian projector (see Section VI)

Finally, we consider a learnable subspace

$$S_{t,k}^{\text{St}} = \text{span}(Q_t),$$

where $Q_t \in \text{St}(k, p)$ is an orthonormal basis on the Stiefel manifold. This variant allows the subspace itself to be optimized rather than defined through a fixed spectral or coordinate-based rule.

V. QUALITY METRICS

In this paper, we plan to investigate the behavior of optimizers by projecting their updates onto low-dimensional subspaces, which naturally leads to the following metrics:

Alignment

$$\chi_S(v) = \frac{\|P_S v\|}{\|v\|}$$

This metric shows what proportion of the vector lies within the selected subspace.

In addition to the instantaneous alignment value, we use an exponential moving average (EMA) of the alignment score:

$$\bar{\chi}_{S,t} = \beta \bar{\chi}_{S,t-1} + (1 - \beta) \chi_S(v_t).$$

Following the protocol of Song et al. (2025), the projection phase starts once the EMA of the alignment score exceeds the threshold 0.95. In our experiments, this switching rule determines the first step from which we replace the full optimizer update by either the projected update $P_{S_t} u_t$ or the orthogonal projected update $(I - P_{S_t}) u_t$.

Loss/Accuracy

As task-level metrics, we report training loss, validation loss, and accuracy. These metrics are used to test whether high alignment with a subspace corresponds to actual learning progress, since a subspace may explain most of the gradient norm while still failing to decrease the loss or improve predictive quality.

Projected descent ratio

Alignment alone does not show whether a subspace is useful for optimization. Therefore, we also measure how much of the predicted one-step loss decrease is preserved after projecting the optimizer update onto a subspace.

Using the second-order Taylor approximation,

$$L(\theta - \eta u) \approx L(\theta) - \eta \langle g, u \rangle + \frac{\eta^2}{2} u^\top H u,$$

where $g = \nabla L(\theta)$, $H = \nabla^2 L(\theta)$, and u is the update direction produced by the optimizer.

For a subspace S with projector P_S , the projected update is

$$u_S = P_S u.$$

We define the projected descent ratio as

$$\rho_S = \frac{\eta \langle g, P_S u \rangle - \frac{\eta^2}{2} (P_S u)^\top H (P_S u)}{\eta \langle g, u \rangle - \frac{\eta^2}{2} u^\top H u}.$$

This metric compares the predicted loss decrease of the projected update with the predicted loss decrease of the original full update.

The interpretation is as follows:

- $\rho_S \approx 1$: the projected update preserves almost all useful loss decrease of the full update, so the subspace is useful for optimization;
- $0 < \rho_S < 1$: the projected update decreases the loss, but less effectively than the full update;
- $\rho_S \approx 0$: the projected update provides almost no useful progress;
- $\rho_S < 0$: the projected update is predicted to increase the loss, so the subspace is harmful for optimization;
- $\rho_S > 1$: the projected update is predicted to decrease the loss even more than the full update.

In our experiments, a subspace is considered useful only if it has a high projected descent ratio and this is confirmed by the actual training loss and validation accuracy. Thus, high alignment $\chi_S(u)$ is not sufficient by itself: a subspace may explain most of the update norm but still fail to produce meaningful loss decrease.

VI. THEORETICAL ANALYSIS AND EXPLANATION

A. SGD in Gradient-Flow regime

Theorem 1 (SGD alignment does not imply projected trainability). *Consider a local quadratic approximation*

$$L(\theta) = \frac{1}{2} x^\top \Lambda x + \frac{1}{2} y^\top M y, \quad \theta - \theta^* = U_k x + U_\perp y,$$

where U_k spans the dominant k -dimensional Hessian eigenspace,

$$H = U_k \Lambda U_k^\top + U_\perp M U_\perp^\top,$$

and the spectrum is separated as

$$\lambda_1 \geq \dots \geq \lambda_k \gg \mu_{k+1} \geq \dots \geq \mu_d \geq 0.$$

Let SGD operate in the small-step, gradient-flow regime,

$$\theta_{t+1} = \theta_t - \eta (\nabla L(\theta_t) + \xi_t), \quad \mathbb{E}[\xi_t \mid \theta_t] = 0,$$

with $0 < \eta \lambda_1 < 2$. Then minibatch noise produces a curvature-amplified gradient response: in the noise-driven

regime the expected gradient energy is biased toward the dominant Hessian subspace. In particular, for isotropic coordinate noise and

$$\Lambda = \lambda I_k, \quad M = \mu I_{d-k},$$

the alignment surrogate satisfies

$$\bar{\chi}_k := \frac{\mathbb{E} \|P_k \nabla L(\theta_t)\|^2}{\mathbb{E} \|\nabla L(\theta_t)\|^2} \approx \frac{k\lambda}{k\lambda + (d-k)\mu}.$$

Hence, if

$$\frac{\lambda}{\mu} \gg \frac{d-k}{k},$$

the full gradient can be almost entirely aligned with the low-dimensional dominant Hessian subspace.

However, this alignment does not imply that training can be restricted to this subspace. If after some time t_0 we replace the dynamics by projected gradient flow

$$\dot{\theta} = -P_k \nabla L(\theta),$$

or by projected SGD

$$\theta_{t+1} = \theta_t - \eta P_k (\nabla L(\theta_t) + \xi_t),$$

then all bulk coordinates are frozen:

$$\dot{y} = 0 \quad \text{or} \quad y_{t+1} = y_t.$$

Consequently, projected training converges only to the projected stationary set

$$\mathcal{M}_{\text{proj}} = \{\theta : P_k \nabla L(\theta) = 0\} = \{\theta^* + U_\perp y : y \in \mathbb{R}^{d-k}\},$$

which is generally not the optimum set of the original loss. In particular, if $y(t_0) \neq 0$, projected gradient flow leaves a residual loss

$$L_\infty = \frac{1}{2} y(t_0)^\top M y(t_0) > 0.$$

Thus, in the gradient-flow regime, SGD may exhibit strong gradient alignment with a low-dimensional Hessian subspace, while optimization restricted to that same subspace can stop prematurely because the remaining bulk coordinates no longer evolve.

Proof of this Theorem is provided in Appendix B

B. Projectors expalnation

a) *Stiefel projector*: The Stiefel projector is updated directly on the manifold. At step t , we first compute the optimizer update direction u_t : the gradients are obtained by backpropagation and then transformed by the optimizer state (for example, momentum or adaptive preconditioning). The Stiefel basis is trained to increase the captured update energy,

$$\max_{Q \in \text{St}(k, p)} \frac{1}{2} \|Q^\top u_t\|^2.$$

Let $a_t = Q_t^\top u_t$. The Euclidean gradient of this objective is $u_t a_t^\top$. We project it onto the tangent space of the Stiefel manifold at Q_t ,

$$\xi_t = u_t a_t^\top - Q_t a_t a_t^\top \in T_{Q_t} \text{St}(k, p),$$

and then return to the manifold by a retraction,

$$Q_{t+1} = R_{Q_t}(\eta_{\text{St}} \xi_t).$$

In our implementation, we use two retractions: the Cayley retraction and the polar projection. For the projection the polar factor is computed approximately using the CANS polynomial iteration.

b) Layerwise momentum subspace (Inspired by Gailore (Zhao et al., 2024) and MoFaSGD (Mahdavinia et al., 2025)): For optimizers with momentum, particularly AdamW, let m_t denote the momentum vector of the first-moment estimate. In addition to treating m_t as a single vector, we consider its layerwise matrix representation. For each layer ℓ , that is not a vector, the corresponding momentum tensor is reshaped into a matrix:

$$M_t^{(\ell)} \in \mathbb{R}^{d_{\text{in}} \times d_{\text{out}}}$$

For convolutional kernels, we flatten the non-output dimensions into the second matrix dimension. We compute the singular value decomposition (SVD)

$$M_t^{(\ell)} = U_t^{(\ell)} \Sigma_t^{(\ell)} \left(V_t^{(\ell)} \right)^\top$$

and select the leading singular vectors according to a retained energy threshold $g \in (0, 1]$. Namely, we choose the smallest k such that

$$\frac{\sum_{i=1}^k \sigma_i^2}{\sum_i \sigma_i^2} \geq q$$

The threshold q controls the rank of the resulting approximation: smaller q gives a lower-dimensional and more aggressive projection, while larger q makes the update closer to the original optimizer update.

Let $U_{t,k}^{(\ell)}, V_{t,k}^{(\ell)}$ be the matrices containing first k left and right singular vectors of $M_t^{(\ell)}$ respectively. Then consider an optimizer update in matrix form $Z_t^{(\ell)}$. We consider the following tangent-space projection:

$$P_{t,k}^M(Z_t) = U_{t,k}^{(\ell)} \left(U_{t,k}^{(\ell)} \right)^\top Z_t^{(\ell)} + Z_t^{(\ell)} V_{t,k}^{(\ell)} \left(V_{t,k}^{(\ell)} \right)^\top - U_{t,k}^{(\ell)} \left(U_{t,k}^{(\ell)} \right)^\top Z_t^{(\ell)} V_{t,k}^{(\ell)} \left(V_{t,k}^{(\ell)} \right)^\top \quad (1)$$

Under our low-rank assumption on the momentum matrix $M_t^{(\ell)}$, this projection directs the dynamics of optimizer into the stable subspace induced by accumulated gradient information. More strictly, this projection preserves all infinitesimal variations around matrices whose row and column subspaces are spanned by $U_{t,k}^{(\ell)}$ and $V_{t,k}^{(\ell)}$ respectively.

c) Adaptive (preconditioned) Hessian subspace: At the t -th basis refresh let $g = \nabla L(\theta) \in \mathbb{R}^p$ be the full-batch gradient. We maintain a second-moment EMA (the same state Adam keeps) and debias it exactly as Adam does,

$$v \leftarrow \beta_2 v + (1 - \beta_2) g^{\odot 2}, \quad \hat{v} = \frac{v}{1 - \beta_2^t}, \quad 0 \leq \beta_2 < 1,$$

and form the diagonal preconditioner

$$P = \text{diag}(\sqrt{\hat{v}} + \varepsilon) \in \mathbb{R}^{p \times p}.$$

Then Adam’s update is

$$u = -P^{-1} \hat{m}$$

, with the first-moment term $\hat{m}$

The change of variables $y = P^{1/2} x$ maps parameter space to the coordinates in which the adaptive optimizer’s step is isotropic. The EMA is self-maintained, so the construction is well defined even under plain SGD.

Preconditioned Hessian and basis.

With $H = \nabla^2 L(\theta)$ define the symmetric preconditioned (whitened) Hessian

$$\tilde{H} = P^{-1/2} H P^{-1/2}.$$

Its top- k eigenvectors $Q = [q_1, \dots, q_k] \in \text{St}(k, p)$ are obtained as the plain Hessian projector; each matrix–vector product costs a single HVP,

$$\tilde{H} w = d \odot H(d \odot w), \quad d = \text{diag}(P^{-1/2}).$$

The columns of Q live in the whitened coordinates y , so the dominant subspace in parameter space is the pull-back

$$S_{t,k}^{\tilde{H}} = P^{-1/2} \text{span}(Q) = \text{span}(P^{-1/2} q_1, \dots, P^{-1/2} q_k).$$

Dom / bulk projection. The update is split with the projector

$$\Pi = P^{-1/2} Q Q^\top P^{1/2},$$

$$\text{dom}(u) = \Pi u, \quad \text{bulk}(u) = (I - \Pi)u.$$

Π is idempotent ($\Pi^2 = \Pi$, since $Q^\top Q = I_k$) but *not* symmetric in the Euclidean inner product. Writing $\langle a, b \rangle_P = a^\top P b$ for the P -inner product, one checks $\langle \Pi a, b \rangle_P = \langle a, \Pi b \rangle_P$, so Π is the orthogonal projector onto $S_{t,k}^{\tilde{H}}$ in the metric P . Consequently dom and bulk are P -orthogonal,

$$\langle \text{dom}(u), \text{bulk}(u) \rangle_P = \text{dom}(u)^\top P \text{bulk}(u) = 0,$$

and the split is the genuine curvature decomposition of the update in the geometry the optimizer actually moves in.

d) Muon-metric (Kronecker) Hessian subspace: This is the two-sided analog of the previous projector: it replaces the *diagonal* metric P by Muon’s per-layer *Kronecker* metric and acts on each weight matrix separately, constraining the dominant subspace to left $\otimes$ right (singular) structure.

Consider a weight matrix $W \in \mathbb{R}^{m \times n}$ and let $G \in \mathbb{R}^{m \times n}$ be the full-batch refresh gradient. Muon’s metric is the Kronecker product

$$\mathcal{M} = L \otimes R, \quad L = (GG^\top)^{1/2} \in \mathbb{R}^{m \times m},$$

$$R = (G^\top G)^{1/2} \in \mathbb{R}^{n \times n}.$$

$$\text{msign}(G) = UV^\top = L^{-1/2} G R^{-1/2},$$

Because the spectral norm is non-Hilbertian, $\mathcal{M}$ is a state-dependent linearization and is rebuilt every refresh from the current G .

Whitened per-layer Hessian and basis.

Let H_ℓ be the per-layer Hessian block, viewed as a symmetric operator $H_\ell[D]: \mathbb{R}^{m \times n} \rightarrow \mathbb{R}^{m \times n}$ (a 2D HVP). The whitened block is

$$\tilde{H}_\ell[D] = L^{-1/2} H_\ell[L^{-1/2} D R^{-1/2}] R^{-1/2},$$

which is $\mathcal{M}^{-1/2} H_\ell \mathcal{M}^{-1/2}$ in vectorized form — the exact two-sided analog of $\tilde{H} = P^{-1/2} H P^{-1/2}$. We extract its dominant rank- r left/right subspaces $\tilde{U} \in \text{St}(m, r)$, $\tilde{V} \in \text{St}(n, r)$ by HOOI (Higher-order orthogonal iteration), i.e. by alternating maximization of the captured curvature

$$\Phi(U, V) = \text{tr}\left((V \otimes U)^\top \tilde{H}_\ell(V \otimes U)\right),$$

exploiting the $m \leftrightarrow m$, $n \leftrightarrow n$ symmetry so that each half-sweep is one small symmetric eigenproblem. The dominant subspace is thus constrained to the Kronecker form $\text{span}(\tilde{V}) \otimes \text{span}(\tilde{U})$ (a rank- r^2 subspace of $\mathbb{R}^{mn}$), rather than an arbitrary r -dimensional one.

Dom / bulk projection.

With the whitened-coordinate projector $D \mapsto \tilde{U} \tilde{U}^\top D \tilde{V} \tilde{V}^\top$, the $\mathcal{M}$ -orthogonal split is

$$\text{dom}(D) = L^{-1/2} \tilde{U} \tilde{U}^\top (L^{1/2} D R^{1/2}) \tilde{V} \tilde{V}^\top R^{-1/2},$$

$$\text{bulk}(D) = D - \text{dom}(D).$$

Reading right to left: whiten $D \mapsto Y = L^{1/2} D R^{1/2}$, project orthogonally in whitened coordinates $Y \mapsto \tilde{U} \tilde{U}^\top Y \tilde{V} \tilde{V}^\top$, then un-whiten. As in the diagonal case this is the projector onto the dominant curvature subspace that is orthogonal in the optimizer’s metric, so dom and bulk are $\mathcal{M}$ -orthogonal.

Spectrum floor.

A finite-batch G is rank-deficient ($\text{rank}(G) \leq \min(m, \text{batch})$), so $R^{-1/4}$ would explode on the null space of G and pull the whitened-curvature subspace into directions the update never visits. Each Gram spectrum is therefore floored at $\varepsilon \lambda_{\max}$ before taking the $\pm 1/4$ powers, which bounds the whitening’s condition number to $\varepsilon^{-1/4}$ while leaving the dominant spectrum ($\lambda > \varepsilon \lambda_{\max}$) Muon-exact. The limit $\varepsilon \rightarrow \infty$ floors the spectrum flat $\Rightarrow$ the metric becomes isotropic $\Rightarrow$ the construction recovers the Euclidean (plain spectral) Hessian projector.

e) *What survives the change of metric.*: All three projectors use the whitened Hessian $\tilde{H} = \mathcal{M}^{-1/2} H \mathcal{M}^{-1/2}$ with an SPD metric $\mathcal{M} \succ 0$, i.e. a congruence of H . Congruence is exactly the property the dom/bulk construction relies on: the split is $\mathcal{M}$ -orthogonal, and rank, inertia, the Gauss–Newton factorization (hence the rank $\leq N(k-1)$ bound) and the class-component dimension $k = \# \text{classes}$ are all metric-invariant (Appendix C). What congruence does *not* preserve

are eigenvalues and eigenvectors, so the two properties that decide whether the subspace is useful are genuinely $\mathcal{M}$ -dependent: *dominance* — whether the $\leq k$ class directions are the *top* eigenvectors of $\tilde{H}$ — and *alignment* — whether the update concentrates in that subspace ($\chi_k \rightarrow 1$) (Gur-Ari et al., 2018; Song et al., 2025). A metric can sharpen or erase the outlier gap, so establishing either needs an explicit hypothesis relating $\mathcal{M}$ to H .

VII. SETTINGS

First, we implemented all neural network architectures considered in Song et al. (2025):

- 1) 3-layer MLP with a width of 200 and tanh activation functions;
- 2) 3-layer CNN with a width of 32 and ReLU activation functions;
- 3) 2-layer Transformer with a hidden dimension of 64 and 8 attention heads

Second, we prepared the following datasets used in Song et al. (2025):

- 1) MNIST-5k: a subset of 5 000 training samples from the MNIST dataset (LeCun, 1998);
- 2) CIFAR10-5k: a subset of 5 000 training samples from the CIFAR-10 dataset (Krizhevsky, 2009);
- 3) SST2-1k: a subset of 1 000 training samples from the SST-2 dataset (Socher et al., 2013).

Third, we added support for a broader set of optimizers. In addition to the optimizers considered in the original paper, our implementation currently supports:

- 1) SGD;
- 2) SGD with momentum;
- 3) Adam;
- 4) AdamW;
- 5) Muon with several polynoms
 - a) CANS (Grishina et al., 2026)
 - b) Polar-Express (Amsel et al., 2026)
 - c) Newton-Schulz (Jordan et al., 2024a)
- 6) forward-gradient methods;
- 7) MeZO;
- 8) SubZero.

Finally, we developed a unified experimental runner. It allows us to run the same experimental protocol for an arbitrary combination of architecture, optimizer, and projector. This makes it possible to compare full updates, projected updates, and orthogonal projected updates in a consistent way across different experimental settings.

VIII. ADAM AND THE ADAPTIVE EDGE OF STABILITY

We train a small transformer on SST-2 with Adam (Kingma et al., 2015) at $\beta_1 = 0$, $\beta_2 = 0.999$, $\varepsilon = 10^{-8}$, so that P is exactly the optimizer’s preconditioner (debaised RMSProp). The basis Q of $\tilde{H} = P^{-1/2}HP^{-1/2}$ is computed by full-dataset Lanczos with $k = 2 = \text{\#classes}$ (Theorem 4). We use two regimes: *measurement-only*, in which χ_k , the stable rank and the spectrum are logged along the unmodified Adam trajectory, and *intervention*, in which the update is replaced by its dom- or bulk-component once the alignment EMA first exceeds 0.88. All curves are single-seed and illustrative.

A. Spectral structure and stable rank

Figure 1 confirms $\lambda_k/\lambda_{k+1}(\tilde{H}) < \lambda_k/\lambda_{k+1}(H)$ at every k : the sharp internal gaps of the raw Hessian (ratios 2.18 at $k=2$ and 5.0 at $k=4$) collapse to near unity after whitening, lifting the stable rank. This is the expected effect of a Fisher-type preconditioner $P \approx \text{diag}(\text{Fisher})^{1/2}$ (Kunstner et al., 2019), which de-concentrates the top of the spectrum. It is consistent with Theorem 3: whitening is a congruence, so it leaves $\text{rank}(H)$ and the $\leq N(k-1)$ bound intact and only redistributes weight within the low-rank block — i.e. dominance, the metric-dependent property.

The raw spectrum reproduces the known outlier-plus-bulk structure (Sagun, Evci, et al., 2018; Ghorbani et al., 2019; Pappan, 2019). The two leading directions match $\dim C \leq k = 2$ (Theorem 4), and the emerging second pair is consistent with the class $\times$ class level of Pappan’s hierarchy (Pappan, 2019; Pappan, 2020). (The first logged step is discarded: the cold debaised second moment makes $\tilde{H}$ transiently ill-scaled.)

B. Update–curvature alignment

Figure 2 establishes alignment for the Adam metric, and Lemma 4 explains the gap. With $\beta_1 = 0$ the update is $u = -P^{-1}g$, so the whitened update is $P^{1/2}u = -P^{-1/2}g = -\tilde{g}$, exactly the whitened gradient, which lies in $\text{col}(\tilde{H}_{\text{GN}})$. Hence χ_k against $\tilde{H}$ measures the whitened-metric instance of the Gur-Ari outlier-alignment mechanism (Gur-Ari et al., 2018; Arous et al., 2024); the Euclidean projector instead asks whether $-P^{-1}g$ lies in the top- k of H , but P^{-1} tilts g out of $\text{col}(H_{\text{GN}})$. The point is not simply that a metric helps: the raw Hessian has the sharper outlier gap yet aligns worse, because Adam does not step in the Euclidean geometry. Alignment is high exactly when the projector’s metric equals the optimizer’s.

Two caveats. First, χ_k is measured in the metric that defines the subspace, so a high value is not itself proof that training happens there — the causal test is the intervention below. Second, both alignments decay substantially as the model approaches interpolation: the residuals vanish, the Gauss–Newton signal that creates the outliers weakens, and the gradient becomes noise. This differs from the persistent SGD alignment of Gur-Ari et al. (2018) and matches the caution of Song et al. (2025).

C. Where the descent lives: dom/bulk intervention

Figure 3 is the causal counterpart of the alignment measurement: the Adam analog of the phenomenon of Gur-Ari et al. (2018) and Song et al. (2025), in which training proceeds in the complement of the dominant subspace. Restricting the update to the top-2 whitened directions does not merely stall training, it reverses it: the loss rises by more than an order of magnitude while accuracy stays at 1.0, so the iterate erodes margin. The complementary bulk update, with the basis frozen at the switch, descends below unprojected Adam at roughly $100\times$ lower per-step cost. The descent-usefulness ρ (loss decrease per unit update norm) confirms the split: $\rho_{\text{dom}} = 0.55 < 1 < \rho_{\text{bulk}} = 2.2$.

The two bulk curves differ only in whether the basis is refrozen at the switch. The metric $\mathcal{M} = P$ is state-dependent: removing the dom-component moves θ , which changes g , P , $\tilde{H}$ and its eigenvectors Q , redefining dom/bulk. Iterating the decomposition while the metric co-adapts closes a feedback loop that self-amplifies into the spikes of Figure 3 — the preconditioner compensation of the adaptive edge of stability (Cohen, Ghorbani, et al., 2023). Freezing $\mathcal{M}$ breaks the loop. Rigorous dom/bulk statements for adaptive optimizers thus require a separation of time-scales between metric adaptation and parameter motion.

D. The dominant subspace is the adaptive edge of stability

The remaining question is what fixes the dominant directions and why they carry no descent. The answer ties the construction to the adaptive edge of stability (AEoS) of Cohen, Ghorbani, et al. (2023): the object AEoS identifies as controlling Adam’s stability, the preconditioned Hessian, is exactly $\tilde{H} = P^{-1/2}HP^{-1/2}$.

a) *Stability threshold.*: In the whitened coordinates $y = P^{1/2}\theta$ of Lemma 4, an Adam step with $\beta_1 = 0$ is, to leading order around a quadratic, gradient descent with curvature $\tilde{H}$ and step size η , whose linear stability requires $\eta \lambda_{\max}(\tilde{H}) < 2$ (Cohen, Kaur, et al., 2021):

$$\lambda_{\max}(\tilde{H}) \lesssim \frac{2}{\eta} \quad (\beta_1 = 0),$$

the $\beta_1 = 0$ instance of the general adaptive threshold $\frac{2(1+\beta_1)}{1-\beta_1} \eta^{-1}$ ($= 38/\eta$ at $\beta_1 = 0.9$ in Cohen, Ghorbani, et al. (2023)). AEoS predicts $\lambda_{\max}(\tilde{H})$ rises to and then hovers below this threshold (Damian et al., 2023). Then the top- k subspace of $\tilde{H}$ — our dom subspace — is by construction at the edge of stability: marginally (un)stable directions that only oscillate, which is why projecting the update onto them produces no descent and erodes margin (Fig. 3), while the sub-threshold bulk still contracts.

b) *Learning-rate sweep.*: We sweep $\eta \in \{0.5, 1, 2, 4\} \times 10^{-3}$ (measurement-only) and read $\lambda_{\max}(\tilde{H})$ at its training-time equilibrium (Fig. 4, Table ??). For the three stable rates the equilibrium scales as $1/\eta$, with $\eta \lambda_{\max}(\tilde{H}) = 1.924 \pm 0.005$, pinned just below the predicted value 2. This is not an artifact of how P is normalized: across the sweep the median $\sqrt{\hat{v}}$ varies by only about $2\times$, far short of the $8\times$ a $1/\eta$

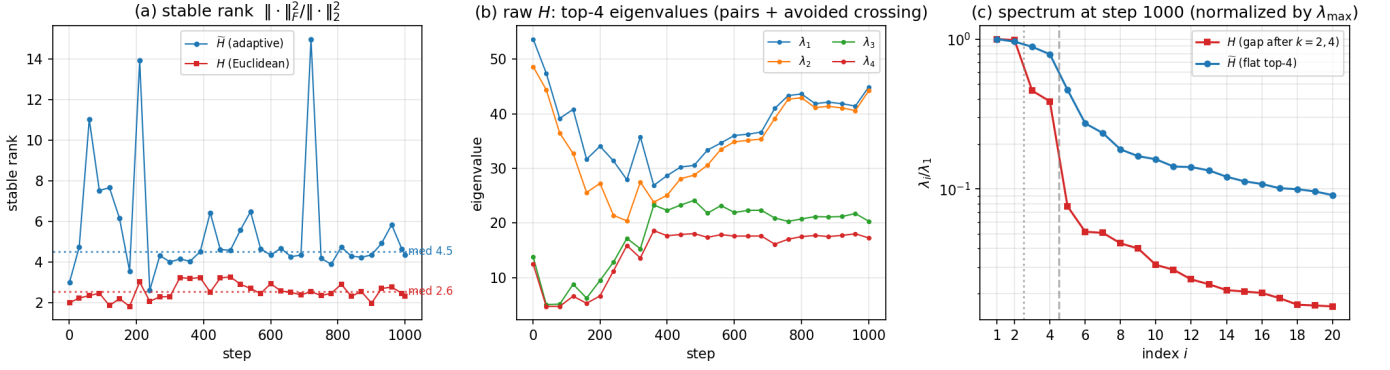


Fig. 1. Whitening flattens the outlier spectrum. (a) Stable rank $sr = \sum_i \lambda_i^2 / \lambda_{\max}^2$ along the Adam trajectory: the raw H stays near 2.6, the whitened $\tilde{H}$ rises to ≈ 4.5 . (b) Top-4 eigenvalues of H : two near-degenerate pairs, with an avoided-crossing reorganization near step 320. (c) Spectra at step 1000 (normalized by $\lambda_{\max}$): H has sharp gaps after $k=2$ and $k=4$; $\tilde{H}$ has a flat top-4.

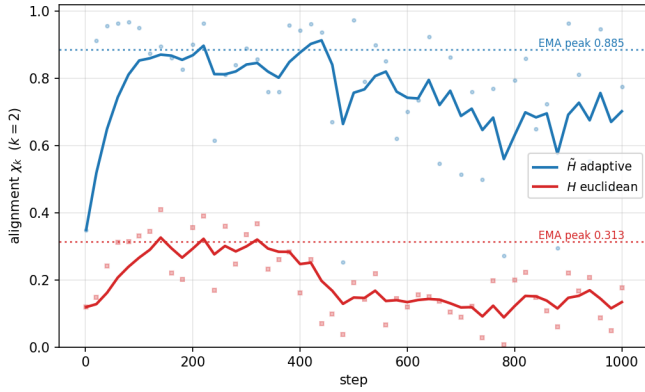


Fig. 2. In its own metric Adam’s update lives in the top- k whitened-curvature subspace; in the Euclidean metric it does not. Alignment χ_k ($k=2$) on the same Adam trajectory (faint: per-step; solid: EMA). Against $\tilde{H}$ it reaches ≥ 0.9 within ~ 30 steps (EMA peak 0.88); against the raw H it peaks at ≈ 0.31 . Both decay toward interpolation.

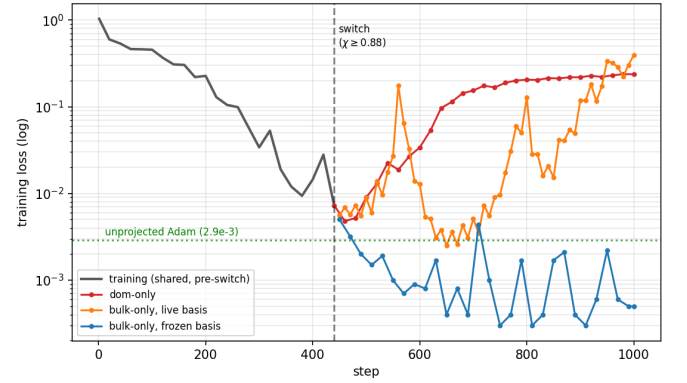


Fig. 3. The dominant whitened-curvature subspace carries no descent; the bulk does. At step 440 the Adam update is replaced by its dom- or bulk-component ($k=2$, P -orthogonal). *dom-only*: the loss rises at accuracy 1.0 (margin erosion). *bulk-only, frozen basis*: smooth descent below unprojected Adam at far lower per-step cost; with a live basis it trains but spikes.

dependence of P would require. The whitening scale is thus essentially fixed, and it is the Hessian that self-organizes to put the preconditioned sharpness at the threshold. The largest rate is past the boundary: the preconditioner can no longer hold sharpness, and the run passes through a catastrophic excursion — the loss rising to chance level — before recovering.

c) *Synthesis.*: Three observations collapse into one. The adaptive metric flattens the outlier gap (Sec. VIII-A); what survives as the top- k of $\tilde{H}$ is not what the loss curvature makes largest, but the k directions the optimizer drives to its stability threshold $2/\eta$ (Sec. VIII-D); and these are exactly where the whitened Adam update concentrates (Sec. VIII-B) yet which carry no descent (Sec. VIII-C). For an adaptive optimizer the small dominant subspace of Gur-Ari et al. (2018) and Song et al. (2025) is the adaptive-edge-of-stability subspace of the preconditioned Hessian. Gradient methods thus do not happen in the top subspace of the loss Hessian, but in the bulk of the Hessian the optimizer’s own geometry induces.

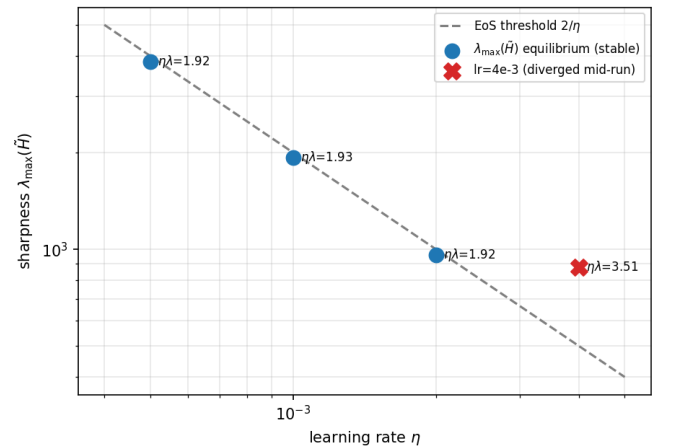


Fig. 4. The whitened sharpness tracks the Adam threshold $2/\eta$. Equilibrium $\lambda_{\max}(\tilde{H})$ versus learning rate (log-log). The three stable rates lie on the dashed $2/\eta$ line; the largest rate ($\eta=4 \times 10^{-3}$, red) is past the stable regime and diverges mid-training before recovering.

IX. MUON AND THE KRONECKER-METRIC HESSIAN SUBSPACE

We now repeat the construction of Section VIII for Muon’s per-layer Kronecker metric $\mathcal{M} = L \otimes R$, with $L = (GG^\top)^{1/2}$ and $R = (G^\top G)^{1/2}$, for which the orthogonalized step $\text{msign}(G) = L^{-1/2}GR^{-1/2}$ is steepest descent (Jordan et al., 2024b). The metric-invariants carry over unchanged: the dom/bulk split is $\mathcal{M}$ -orthogonal (Lemma 4), the rank and the $\leq N(k-1)$ bound are preserved (Theorem 3), and the candidate dimension $k = \# \text{classes}$ is metric-invariant (Theorem 4). As before, only dominance and alignment stay metric-dependent and must be measured.

The Kronecker metric adds one feature the diagonal case did not have. The two-sided whitening $\mathcal{M}^{-1/2} = (GG^\top)^{-1/4} \otimes (G^\top G)^{-1/4}$ is singular on the null space of the finite-batch gradient G : a rank-deficient G makes $R^{-1/4}$ blow up exactly where the update has no support. The metric must therefore be regularized by a spectrum floor ε . We show that alignment is non-monotone in ε , with an interior optimum, so the unregularized Muon metric is in fact the worst probe.

We train the three-layer MLP `mnist_mlp3` with pure Muon (CANS polynomial), learning rate 0.02, no momentum, seed 42, for 1400 steps. The basis is rebuilt on the full dataset at every refresh; the dominant left/right factors $\tilde{U}, \tilde{V}$ of the whitened block $\tilde{H}_\ell = \mathcal{M}^{-1/2}H_\ell\mathcal{M}^{-1/2}$ are extracted by HOOI with $k = 10 = \# \text{classes}$ (Theorem 4). We report χ_k in parameter space, so the number is comparable to the Euclidean spectral projector. As in the diagonal section we use two regimes: *measurement-only*, in which χ_k , the stable rank and the spectrum are logged along the unmodified Muon trajectory; and *intervention*, in which the update is replaced by its dom- or bulk-component once the alignment EMA first crosses a threshold. All curves are single-seed and illustrative.

A. Alignment is metric-dependent and the metric must be regularized

a) Euclidean vs. Muon, same iterates.: Read along one and the same pure-Muon trajectory, the update aligns with the top- k Euclidean Hessian subspace only at $\chi_k \approx 0.20$, but with the top- k of the whitened Muon-metric Hessian at up to $\chi_k \approx 0.80$, at the optimal spectrum floor $\varepsilon^* \approx 0.03$ (Fig. 5). Lemma 4 explains the gap. Muon’s update is the orthogonal factor $u = UV^\top$, whose singular spectrum is flat, so it spreads its energy across all directions and aligns poorly with any concentrated Euclidean top- k . Whitening undoes exactly this flattening,

$$\mathcal{M}^{1/2}u = L^{1/2}(L^{-1/2}GR^{-1/2})R^{1/2} = G,$$

so in its own metric the construction asks whether the gradient G is captured by the top- k of $\tilde{H}$ — the Kronecker instance of the outlier-alignment mechanism (Gur-Ari et al., 2018), and the two-sided analog of the diagonal identity of §VIII-B.

b) The spectrum-floor sweep.: Sweeping ε from the Euclidean limit to the unregularized limit traces alignment as an inverted-U with three regimes (Fig. 6). For a large floor (the Euclidean limit) the floor sits above the whole Gram

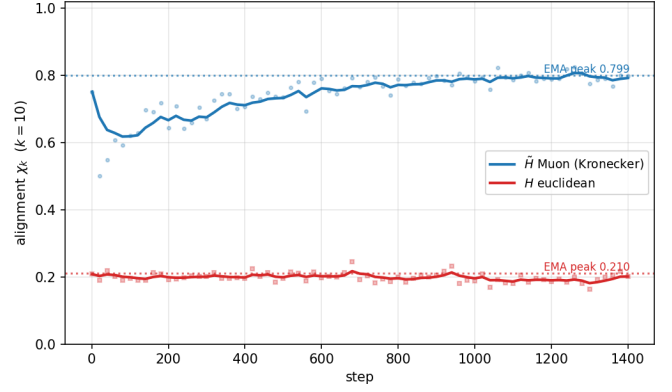


Fig. 5. Muon’s update lives in the top- k Muon-metric (Kronecker) Hessian subspace, not the Euclidean one. Alignment χ_k ($k=10$) of the pure-Muon update along training, against the top- k subspace of each metric’s Hessian, with the Muon metric read at its alignment-optimal spectrum floor.

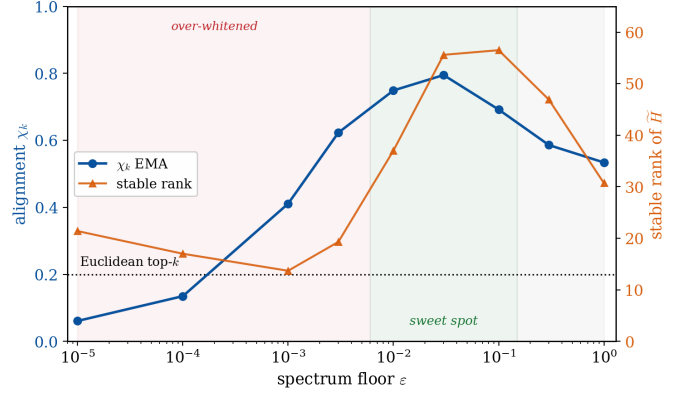


Fig. 6. Alignment in the Muon metric is non-monotone in the spectrum floor ε . Measurement-only, same pure-Muon iterates ($k=10$). Blue: χ_k and its EMA; orange: stable rank of $\tilde{H}$; dotted: the Euclidean top- k baseline. Whitening is sharpest at an intermediate $\varepsilon^* \approx 0.03$; near-Euclidean above it, over-whitened below, where alignment collapses.

spectrum, the metric becomes isotropic, and the projector reduces to the Kronecker-constrained top- k of the Euclidean Hessian. Moderate whitening ($\varepsilon \in [0.01, 0.1]$) re-concentrates the update toward G and alignment peaks at $\varepsilon^* \approx 0.03$. As the floor vanishes (full Muon), alignment collapses below even the Euclidean value.

The collapse has a clean explanation. Writing $G = USV^\top$, unregularized whitening scales the direction with singular value σ by $\sigma^{-1/2}$, so as $\varepsilon \rightarrow 0$ the largest entries of $\tilde{H}$ come from the near-null block of G , and the top eigenvectors $\tilde{U}, \tilde{V}$ migrate into the singular subspaces with $\sigma \approx 0$. But the update $u = UV^\top$ lives in the large- σ block, so $\tilde{U} \perp U$ and $\chi_k \rightarrow 0$: the dominant whitened-curvature subspace flees into directions the optimizer never visits. The floor caps the amplification at $\varepsilon^{-1/4}$ per side and keeps the dominant subspace inside the update’s support; the optimal ε^* trades recovering G ’s geometry against over-amplifying $\ker G$.

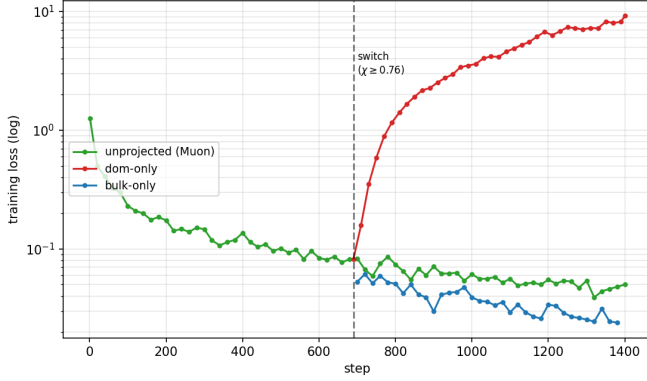


Fig. 7. The dominant whitened-curvature subspace carries no descent; the bulk does, and the effect scales with alignment. The Muon update is replaced by its dom- or bulk-component once the alignment EMA crosses a threshold. (a) Training loss (log): dom (red) reverses training, while bulk (green) descends below unprojected Muon (dashed). (b) Accuracy: dom collapses toward chance; bulk stays near 1.

c) Spectrum.: The whitened spectrum shows no clean outlier gap. Along training the stable rank of $\tilde{H}$ oscillates in $[30, 60]$, well above the Euclidean ≈ 16 on the same iterates: the Kronecker whitening lifts the bulk and de-concentrates the top, the matrix analog of the Fisher-type flattening seen for the diagonal metric (Papayan, 2019). The dimension that is pinned, $k = \# \text{classes}$, is the metric-invariant of Theorem 4; what the Muon metric blurs is dominance, exactly as Section VI allows.

B. Where the descent lives: dom/bulk intervention

Figure 7 is the causal counterpart of the alignment measurement, the Muon analog of the “training happens in the complement” phenomenon of Song et al. (2025). Restricting the update to the top- k whitened directions (dom) does not merely stall training, it reverses it: the loss climbs by orders of magnitude and accuracy falls toward chance. The complementary bulk update descends below the loss unprojected Muon reaches, and the descent-usefulness orders the two the same way, $\rho_{\text{dom}} < 1 < \rho_{\text{bulk}}$.

a) Dose-response.: The two floors tie the correlational alignment metric to the causal intervention: the more aligned the update, the more destructive the dom projection. The more strongly aligned floor drives the loss markedly higher, and accuracy correspondingly lower, than the weaker-aligned one. That the alignment predicts the severity of the intervention is the main evidence that the dominant subspace is mechanistically real and not a measurement artifact. The dom directions are the sharpest in the Muon metric, so a fixed Muon step is super-critical there — a spectral-geometry echo of the adaptive edge of stability of §VIII-D — while the sub-critical bulk still contracts.

b) Euclidean control.: Repeating the intervention with the Euclidean Hessian top- k breaks the clean separation, the sharpest evidence that the metric is necessary rather than decorative. Because the Euclidean alignment stays low, the switch fires only under a much weaker threshold; once applied,

the loss increases under both dom and bulk [exp. 41]. Neither half is “where training happens”. Only in the optimizer’s own (regularized) metric does the dom/bulk split separate useless from useful directions.

C. Discussion

For Muon the three statements collapse into one, with the regularization caveat the diagonal case did not have. The Kronecker metric flattens the outlier gap (no clean SGD-type spectrum); the surviving dominant subspace is captured by the update only when the metric is read in the right, regularized geometry — alignment is largest at an intermediate floor and collapses in the unregularized limit, where the dominant whitened-curvature directions escape into the gradient’s null space; and those dominant directions carry no descent, while the bulk carries all of it and slightly more than the full step. The dose-response between alignment and the size of the dom catastrophe, together with the Euclidean control in which the split scrambles, show that the subspace is mechanistic.

This is the Muon reading of “gradient methods do not happen where we expected”: not in the top subspace of the loss Hessian, and not in the unregularized Muon metric either, but in the bulk of the Hessian induced by the optimizer’s own geometry once that geometry is correctly regularized.

X. UPDATE-SPACE REGULARIZATION VIA TANGENT PROJECTIONS

A. Projection as update-space regularization

Let P be an orthogonal projector, then we define

$$A_\alpha = P + \alpha Q, \quad (2)$$

where $Q := I - P$ is a complementary projector to P . Then consider projected optimizer update:

$$\Delta_\alpha = A_\alpha Z$$

The operator A_α has eigenvalue 1 on the projected subspace and eigenvalue α on the orthogonal complement. Hence, the method acts as an anisotropic filter: update components inside the selected subspace are preserved, while components outside the subspace are suppressed. Equivalently, Δ_α is the solution of the regularized update problem:

$$\Delta_\alpha = \arg \min_{\Delta} \left\{ \frac{1}{2} \|\Delta - Z\|_F^2 + \lambda \|Q\Delta\|_F^2 \right\}$$

Since P and Q are projectors on orthogonal subspaces we have

$$\Delta = P\Delta + Q\Delta, \quad Z = PZ + QZ$$

Then the problem takes the form

$$\min_{\Delta} \left\{ \frac{1}{2} \|P\Delta - PZ\|_F^2 + \frac{1}{2} \|Q\Delta - QZ\|_F^2 + \lambda \|Q\Delta\|_F^2 \right\}$$

And has a solution

$$\Delta_\alpha = PZ + \frac{1}{1 + 2\lambda} QZ \implies \alpha = \frac{1}{1 + 2\lambda}$$

Thus, the update projection method can be interpreted as regularizing the optimizer update rather than the model parameters. Among all updates close to the base optimizer update Z , it selects one with a smaller component outside the selected tangent subspace.

B. Random and norm-corrected controls

To understand whether the regularization effect comes from the specific projection subspace or merely from restricting the update to a lower-dimensional tangent subspace, we consider two additional projections.

First is random tangent projections. In this case, we keep the same layerwise rank k as in momentum-induced construction, but replace $U_{t,k}^{(\ell)}, V_{t,k}^{(\ell)}$ by random orthonormal matrices $\tilde{U}_{t,k}^{(\ell)}, \tilde{V}_{t,k}^{(\ell)}$ of the same dimension. The projection has the same form as (1) but with these matrices.

Second we consider a norm corrected projection:

$$P_{t,k}^{M_{nc}}(Z_t) = \frac{\|Z_t\|_F}{\|P_{t,k}^M(Z_t)\|_F + \varepsilon} P_{t,k}^M(Z_t) \quad (3)$$

This control separates the directional effect of the projection from the trivial shrinkage of the update norm.

C. Controlling regularization via energy threshold

We first study how the retained momentum energy controls the strength of the projection, varying the energy threshold q defined above:

$$q \in \{0.5, 0.6, 0.7, 0.8, 0.9, 0.95\}$$

The results are shown in Figure 8.

The energy threshold has a clear effect on the optimization-generalization trade-off. For smaller values of q , the tangent subspace is lower-dimensional and the applied update is more strongly filtered. This slows down training: the projected methods with lower q have noticeably larger training loss than the AdamW baseline. At the same time, they achieve lower validation loss. In particular, the most aggressive settings, $q = 0.5$ and $q = 0.6$, give the best validation curves, while the baseline AdamW continues to decrease the training loss but overfits more strongly after the projection switch point.

An important observation is that the dependence on q is nonlinear. A moderate increase of the retained singular-value energy can substantially increase the dimension of the tangent space. For example, the transition from $q = 0.8$ to $q = 0.9$ adds a much larger fraction of the parameter space than the transition between lower thresholds. Thus, the energy threshold directly controls the strength of update-space regularization.

Overall, Figure 8 supports the interpretation of q as a regularization parameter. Lower q removes a larger part of the AdamW update, leading to worse training optimization but better validation performance. Higher q weakens the filtering and recovers behavior closer to the base optimizer.

D. Correcting projected update norm

Projection changes not only the direction of the AdamW update, but also its norm. To check whether the improvement is simply caused by a smaller effective step size, we compare the projected update with its norm-corrected version (3). This correction preserves the direction of the projected update but restores its Frobenius norm to the norm of the original AdamW update.

Figure 9 shows that norm correction has almost no visible effect on the behavior of the momentum-induced projection. Both projected variants have higher training loss than the AdamW baseline, but both keep the validation loss lower after the projection is switched on. Moreover, the validation curves of the raw projected update and the norm-corrected projected update are nearly identical.

This suggests that the improvement is not explained solely by a reduction of the update norm. Instead, the main effect comes from changing the update direction: the tangent projection filters the AdamW update in a way that worsens training optimization but improves validation performance.

An auxiliary result for training and validation accuracy is presented in the Appendix (Figure 10).

E. Randomizing tangent projections

We next compare momentum-induced tangent subspaces with random tangent subspaces of the same effective rank as

described in Sec. X-B. Figure 11 shows the final validation performance of these variants. Both momentum-induced and random tangent projections improve over the AdamW baseline. This confirms the projection regularization described in the Sec. X-A.

At the same time, the random variants behave differently under norm correction. Raw random projections achieve the best validation loss and accuracy in this comparison, but their norm-corrected versions lose a substantial part of the gain. This indicates that random tangent projections benefit strongly from update shrinkage or stochastic filtering. In contrast, the momentum-induced projection remains competitive under norm correction, as shown in Sec. X-D.

The refresh frequency has only a minor effect: using $T_{\text{upd}} = 100$ gives performance close to updating the projection basis every step. This is important computationally, since it suggests that the tangent subspace does not need to be recomputed at every optimization step.

Overall, this experiment shows that the projection effect is broader than momentum alignment alone. Random tangent subspaces provide a strong baseline and demonstrate that a substantial part of the improvement comes from low-dimensional update filtering. The full numerical comparison is reported in Appendix (Table I).

F. Residual interpolation

To study the role of the component removed by the projection, we also decompose the update as

$$Z_t = P_{t,k}^M Z_t + (I - P_{t,k}^M) Z_t$$

In our layerwise momentum projections setting we call $P_{t,k}^M Z_t$ the *dominant momentum component* and $(I - P_{t,k}^M) Z_t$ the *bulk momentum component*. Then we consider interpolated projector as in (2):

$$P_{t,k}^{M_\alpha} = P_{t,k}^M + \alpha(I - P_{t,k}^M), \quad \alpha \in [0, 1]$$

This interpolation allows us to test the importance of removed component on model training and generalization.

The results in Appendix (Figure 12) show that adding the residual component consistently moves the method back toward the AdamW baseline. For both momentum-induced and random tangent subspaces, the best validation loss is achieved at $\alpha = 0$. Increasing α worsens validation loss and decreases validation accuracy, even though it returns more of the original AdamW update. This supports the interpretation that the component orthogonal to the tangent subspace is useful for training-set fitting but less useful for generalization in this setting.

This ablation also connects directly to the regularization view from Sec. X-A: α controls how strongly the update is penalized outside the selected tangent subspace. Smaller α gives stronger update-space regularization, while larger α weakens the filtering and recovers behavior closer to AdamW.

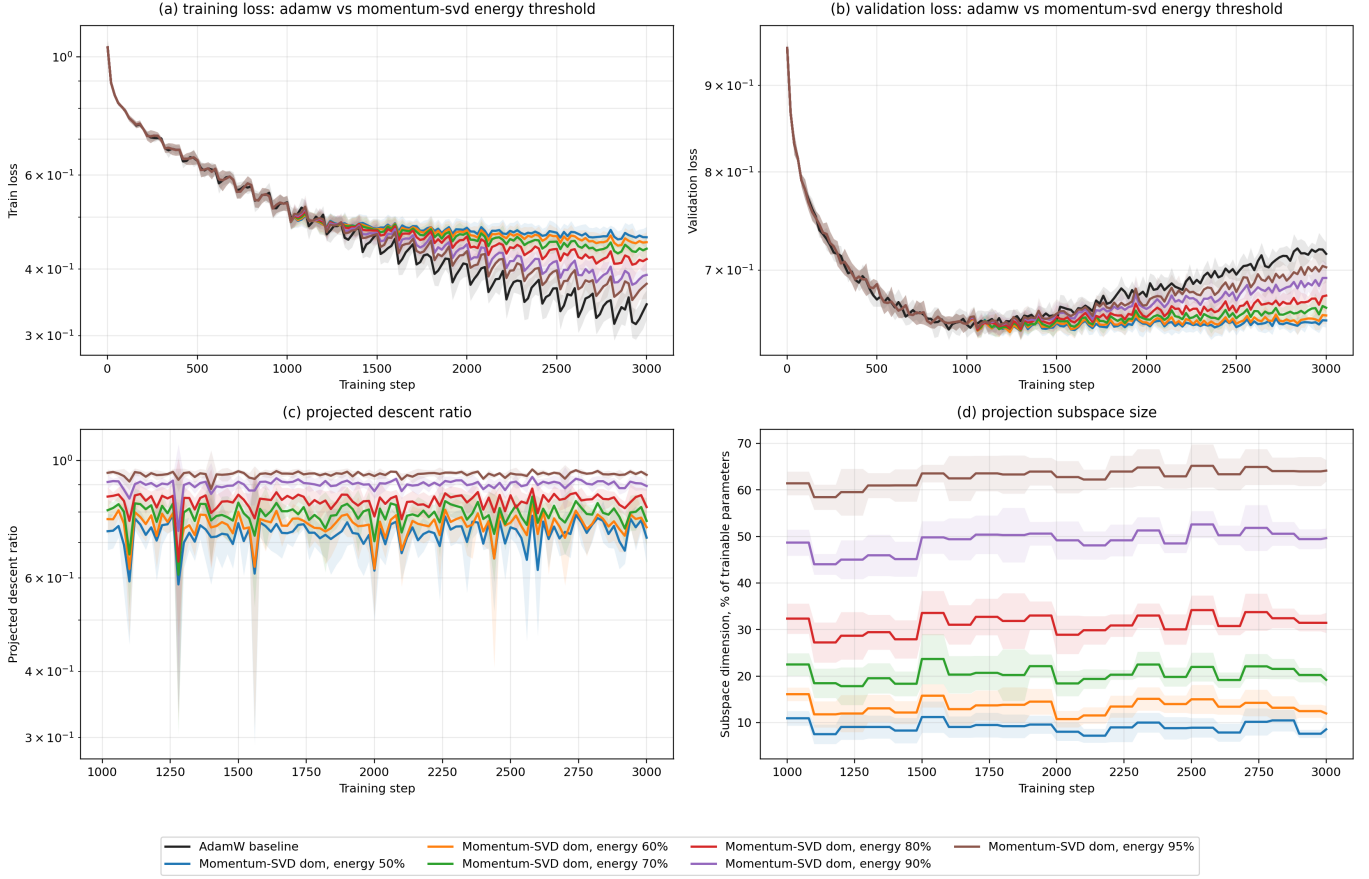


Fig. 8. **Varying energy threshold q controls regularization of AdamW.** (a-b) Training and validation losses: lower q increases train loss but decreases validation loss. (c) Projected descent ratio ρ_S : the usefulness of the subspace for reducing training loss directly depends on q . (d) Projection subspace size depends nonlinearly on q : for $q = 0.5$ it is about 10% of the parameter-space dimension, while for $q = 0.95$ it exceeds 60%.

TABLE I
FINAL METRICS FOR ADAMW MOMENTUM AND RANDOM DOMINANT PROJECTIONS. VALUES ARE MEAN $\pm$ STANDARD DEVIATION OVER FIVE SEEDS.

Method	Train loss	Val. loss	Train acc.	Val. acc.	Subspace (%)
AdamW	0.343 $\pm$ 0.022	0.716 $\pm$ 0.011	0.920 $\pm$ 0.042	0.565 $\pm$ 0.005	–
Momentum energy, $T_{\text{upd}} = 100$	0.417 $\pm$ 0.020	0.676 $\pm$ 0.016	0.892 $\pm$ 0.046	0.572 $\pm$ 0.012	31.4 $\pm$ 2.2
Random energy, $T_{\text{upd}} = 100$	0.394 $\pm$ 0.024	0.668 $\pm$ 0.011	0.920 $\pm$ 0.058	0.584 $\pm$ 0.008	32.3 $\pm$ 4.2
Random energy + norm, $T_{\text{upd}} = 100$	0.383 $\pm$ 0.025	0.687 $\pm$ 0.016	0.932 $\pm$ 0.059	0.578 $\pm$ 0.010	30.2 $\pm$ 4.9
Momentum energy, $T_{\text{upd}} = 1$	0.421 $\pm$ 0.017	0.673 $\pm$ 0.010	0.836 $\pm$ 0.043	0.576 $\pm$ 0.013	30.5 $\pm$ 4.1
Random energy, $T_{\text{upd}} = 1$	0.396 $\pm$ 0.013	0.666 $\pm$ 0.009	0.840 $\pm$ 0.040	0.586 $\pm$ 0.010	26.6 $\pm$ 6.3
Random energy + norm, $T_{\text{upd}} = 1$	0.372 $\pm$ 0.012	0.690 $\pm$ 0.013	0.888 $\pm$ 0.041	0.576 $\pm$ 0.013	28.0 $\pm$ 4.6

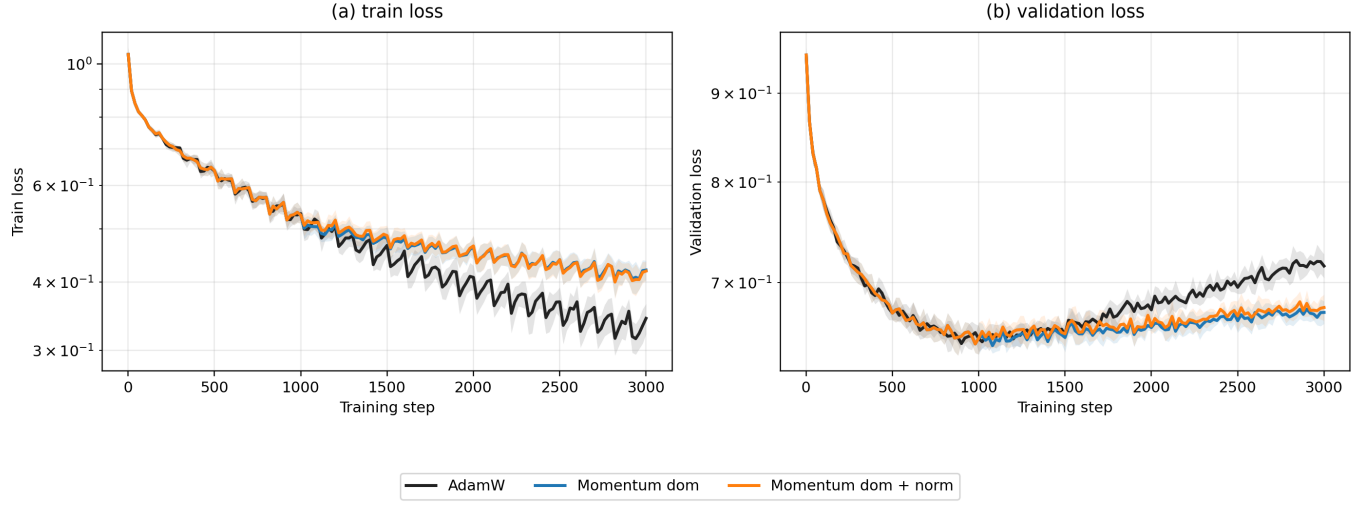


Fig. 9. **Norm correction for AdamW momentum dom subspace, $q = 0.8$, $T_{\text{upd}} = 1$** (a-b) Training and validation losses. The norm-corrected projection closely follows the raw projected update. This indicates that the regularization effect is not caused only by shrinking the update norm.

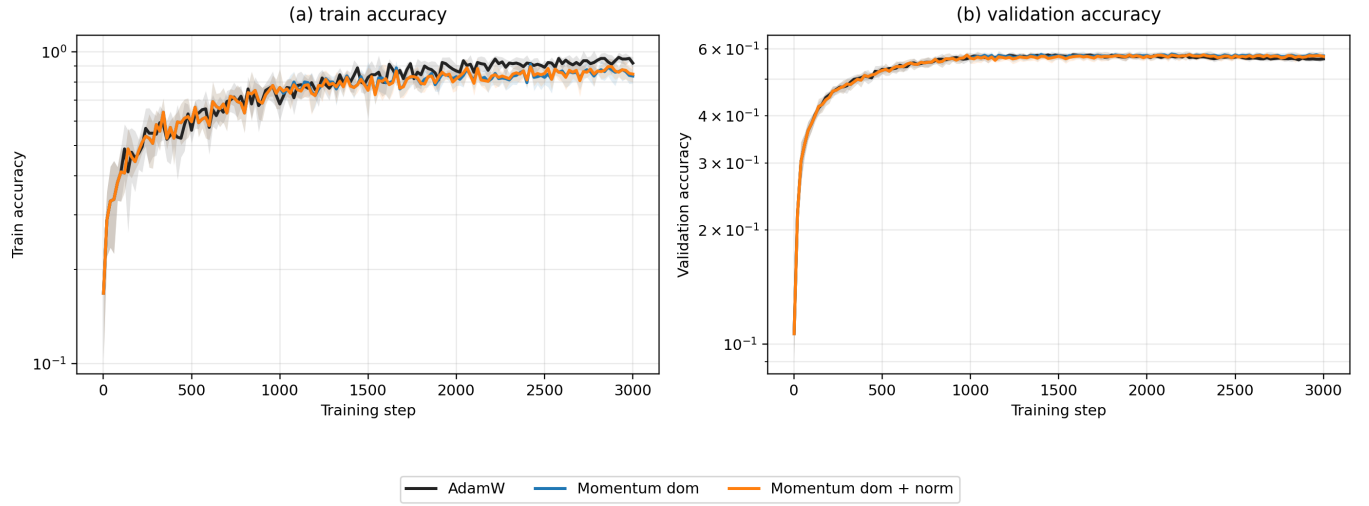


Fig. 10. **Norm correction for AdamW momentum dom subspace, $q = 0.8$, $T_{\text{upd}} = 1$** (a-b) Training and validation accuracy. The norm-corrected projection closely follows the raw projected update. This indicates that the regularization effect is not caused only by shrinking the update norm.

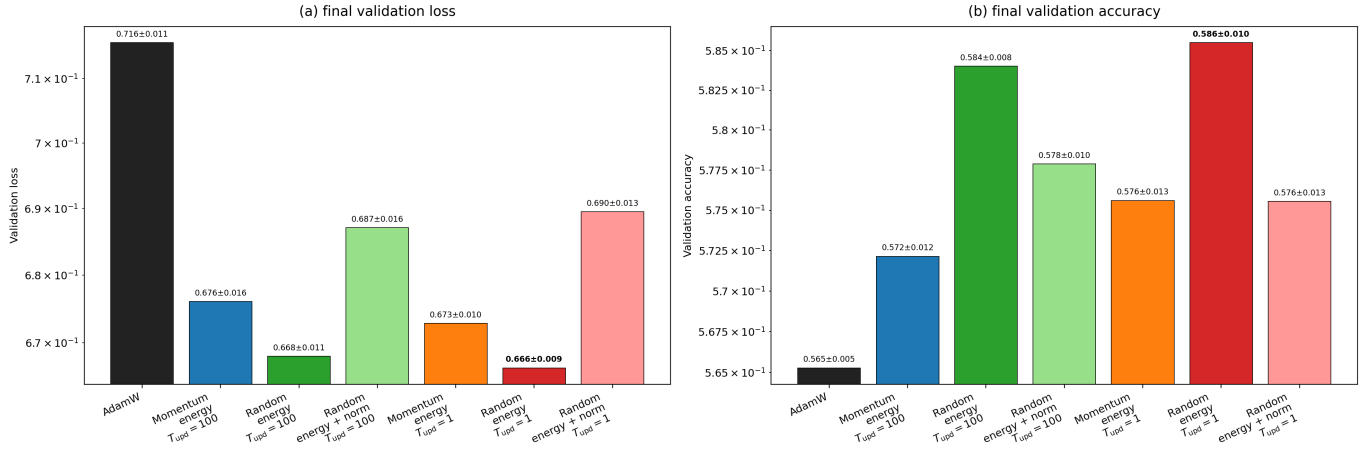


Fig. 11. **Momentum and random dom comparison, $q = 0.8$, different T_{upd}** (a-b) Final validation loss and accuracy. Both momentum and random tangent projections improve over the AdamW baseline, while random projections give the best raw validation performance. However, norm correction substantially reduces the gain of random projections, suggesting that their effect is partly due to update shrinkage or stochastic filtering. Updating the projection basis every 100 steps performs quite similarly to updating it every step.

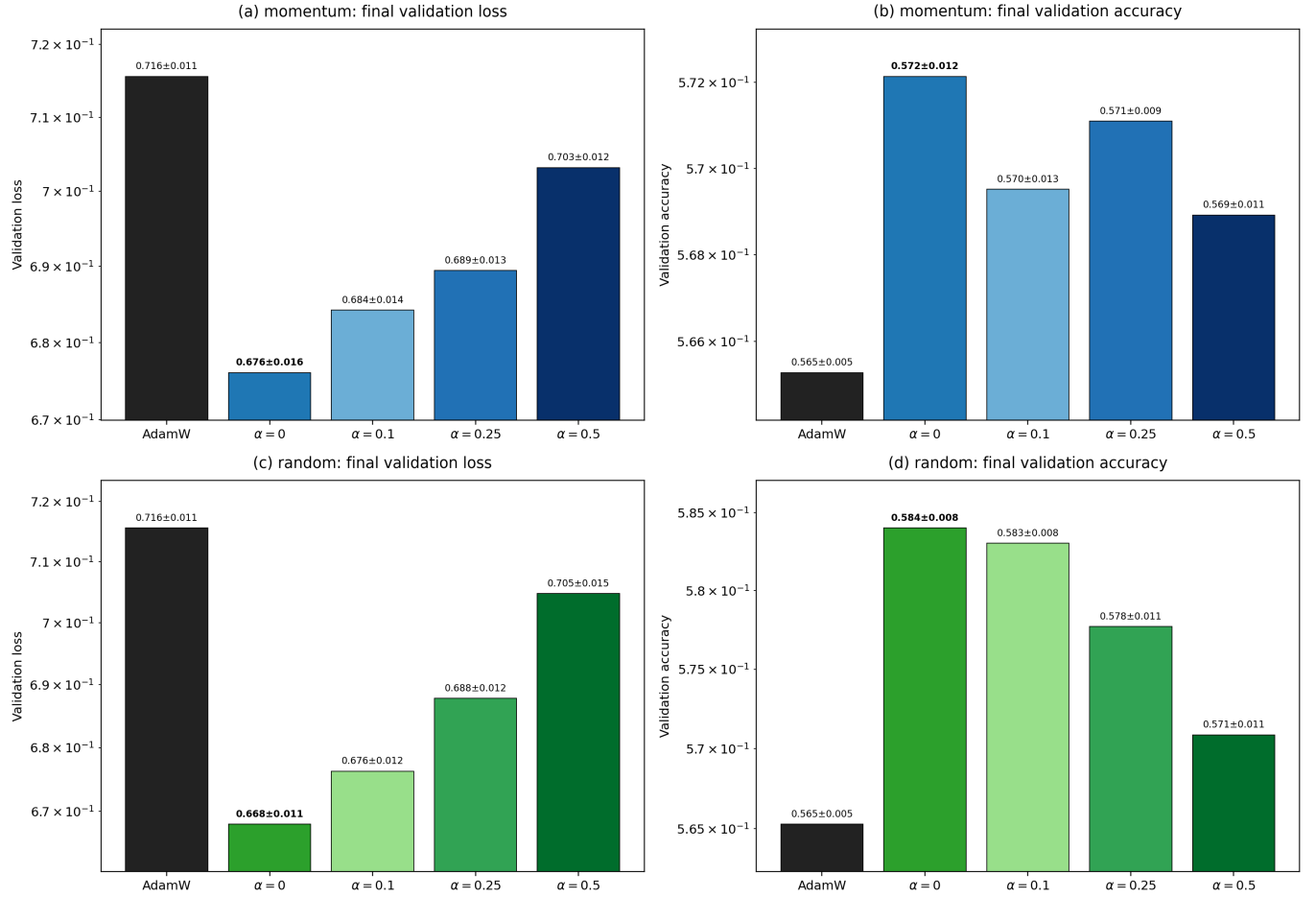


Fig. 12. **Residual interpolation, $q = 0.5$, different α** (a-b) Momentum dom validation loss and accuracy. (c-d) Random dom validation loss and accuracy. The case $\alpha = 0$ corresponds to hard projection, while $\alpha = 1$ recovers the AdamW update. For both momentum-induced and random tangent subspaces, increasing α worsens validation loss and reduces validation accuracy, indicating that the orthogonal residual component helps recover AdamW-like behavior but hurts generalization. Values above bars show mean $\pm$ standard deviation over seeds.

XI. A LEARNABLE STIEFEL SUBSPACE CAPTURES THE UPDATE ONLINE

Unlike the spectral and coordinate subspaces, which are fixed by an external rule, the Stiefel projector *learns* its subspace from the optimizer’s own update stream. We use the CNN on CIFAR10-5k with plain SGD. At each step the basis $Q_t \in \text{St}(k, p)$ is moved one retraction step in the direction that increases the captured update energy $\frac{1}{2}\|Q_t^\top u_t\|^2$ (the Stiefel projector of Section VI-B, Cayley retraction), and the SGD step is replaced by its dominant projected component $P_{S_t} u_t = Q_t Q_t^\top u_t$. The diagnostic is the alignment EMA $\bar{\chi}_{S,t}$ of the update with the *learned* subspace. We compare this run against an unmodified SGD baseline that uses the full update (Figure 13).

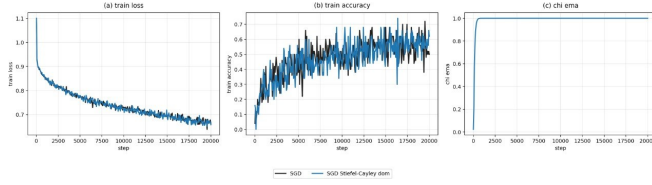


Fig. 13. **Learnable Stiefel subspace on the CNN/CIFAR10-5k setting.** Unmodified SGD (black) versus SGD whose update is projected onto the learned k -dimensional Stiefel subspace with a Cayley retraction (“SGD Stiefel-Cayley dom”, blue). (a) training loss and (b) training accuracy: the projected run tracks the full-update baseline throughout the 2×10^4 steps, with no measurable gap. (c) alignment EMA $\bar{\chi}_{S,t}$ of the update with the learned subspace, which rises from ≈ 0 to ≈ 1 within a few hundred steps and stays there.

Two things happen, and only one of them is by construction. First, the basis locks onto the update almost immediately: $\bar{\chi}_{S,t}$ climbs to ≈ 1 in a few hundred steps and is then pinned there for the rest of training (Figure 13c). On its own this is unsurprising — the basis is optimized precisely to maximize $\|Q_t^\top u_t\|$, so a high captured energy is what the objective rewards. What it does establish is that a single low-dimensional Q_t , carried forward and only nudged each step, suffices to absorb essentially the whole update; the subspace the SGD step lives in is both low-rank and slowly varying, otherwise a one-step retraction could not keep $\bar{\chi}_{S,t}$ saturated.

Second, and this is the part that is *not* guaranteed by the objective, the restriction is free: replacing every SGD step by its projection onto the learned k directions leaves both the training loss and the training accuracy indistinguishable from full-update SGD over the entire run (Figure 13a,b). High alignment alone never implies this — as noted in Section VI, a subspace can carry most of the update norm yet still fail to move the loss. Here the two coincide: the discarded component $(I - P_{S_t})u_t$ carries almost no update energy *and* almost no descent, so the optimizer trained inside the learned subspace follows the same trajectory as the unconstrained one. This is the learnable-subspace counterpart of the fixed-subspace result: the gradient method really does happen in a low-dimensional subspace, and here that subspace is discovered online rather than supplied by the Hessian.

a) *The CANS polar retraction violates the manifold constraint.* We repeat the run but swap the Cayley retraction for the approximate polar projection, whose orthogonal factor is computed by the CANS polynomial iteration (Section VI-B). The alignment diagnostic now leaves its admissible range: $\bar{\chi}_{S,t}$ rises *past* 1 and then oscillates around ≈ 1.0 with peaks near 1.15 (Figure 14). For a genuine orthogonal projector this cannot happen — $\chi_S(v) = \|P_S v\|/\|v\| \leq 1$ because orthogonal projection is non-expansive. A measured value above 1 can therefore only mean that $P_{S_t} = Q_t Q_t^\top$ has stopped being an orthogonal projector: the columns of Q_t have lost their orthonormality and Q_t has drifted off $\text{St}(k, p)$.

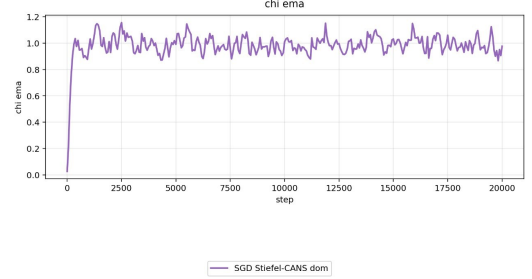


Fig. 14. **The CANS polar retraction breaks orthonormality.** Same CNN/CIFAR10-5k Stiefel run as Figure 13 but with the approximate polar (CANS) retraction instead of Cayley. The alignment EMA $\bar{\chi}_{S,t}$ climbs past the admissible ceiling 1 and oscillates around 1.0 with peaks near 1.15 — a value impossible for a true orthogonal projector, signalling that Q_t no longer satisfies $Q_t^\top Q_t = I_k$.

We attribute this to the inexactness of the CANS polar factor. The polynomial only *approximates* the orthogonal factor of the retracted point, so each step leaves a small residual $Q_t^\top Q_t - I_k \neq 0$ that the iteration never corrects. These residuals accumulate, $\|Q_t\|_2$ creeps above 1, and consequently $\|Q_t Q_t^\top u\|$ can exceed $\|u\|$. The Cayley retraction of Figure 13 keeps Q_t exactly on the manifold and pins χ at 1; the CANS approximation does not. The overshoot is thus a direct, quantitative signature of a broken manifold constraint rather than anything about the data, and it makes the saturated alignment in Figure 13c the more meaningful check — there the retraction was exact and the ceiling was respected.

XII. SWA REMOVES THE DOMINANT-SUBSPACE ALIGNMENT

We reuse the SGD/MLP/MNIST-5k reproduction setting. Instead of projecting the update we add a passive diagnostic. From step $T_{\text{swa}} = 790$, the first step at which the alignment EMA $\bar{\chi}_k$ crosses the 0.95 threshold of Song et al. (2025), we accumulate the SWA average (Izmailov et al., 2018)

$$\bar{\theta}_t = \frac{1}{t - T_{\text{swa}} + 1} \sum_{\tau=T_{\text{swa}}}^t \theta_\tau$$

in a side buffer, while the live iterate θ_t keeps training with unmodified SGD. For $t \geq T_{\text{swa}}$ all weight-dependent quantities — the loss, the alignment, and the top- k eigenvectors — are recomputed at $\bar{\theta}_t$; a companion run evaluates them at θ_t . The buffer never feeds back into the optimizer, so the two runs share one trajectory and coincide up to the switch: the experiment only asks in which subspace the gradient of the averaged model lies (Figure 15).

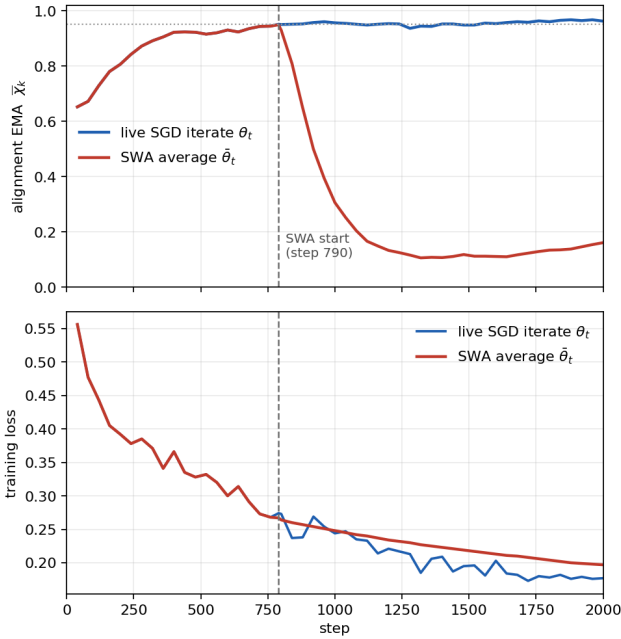


Fig. 15. **Live vs. averaged alignment across the switch.** SWA diagnostic for the MLP on MNIST-5k with SGD, evaluated on the live iterate θ_t (blue) and the SWA average $\bar{\theta}_t$ (red); the dashed line marks T_{swa} . *Top*: alignment EMA $\bar{\chi}_k$ with the top- k Hessian subspace, which reaches the threshold before the switch; afterwards the live iterate stays aligned and the averaged gradient does not. *Bottom*: training loss.

Up to the switch the two diagnostics are identical and reproduce the alignment phase of Gur-Ari et al. (2018) and Song et al. (2025), $\bar{\chi}_k$ reaching the threshold. Afterwards they separate: the live iterate stays aligned, while on the average $\bar{\chi}_k$ collapses smoothly to ≈ 0.1 over a few hundred steps — the gradient of the averaged model has almost entirely left the top- k subspace for the bulk. The collapse is gradual rather than a jump, since at the switch the average still equals the live weight and the gap grows only as the buffer fills; the averaged

training loss falls smoothly and stays marginally above the live one.

a) Why.: The collapse follows from the quadratic model of Section VI. In the sharp coordinates x (curvatures λ_i) SGD settles into a zero-mean stationary oscillation, whereas the bulk coordinates y ($\mu_j \ll \lambda_i$) still carry a slowly decaying nonzero mean drift $m_t = \mathbb{E}[y_t]$; the live alignment is precisely the curvature-amplified gradient energy of those sharp oscillations. Averaging acts oppositely on the two blocks: in the sharp directions it averages a zero-mean sequence, so $\bar{x}_t \rightarrow 0$, while in the bulk it averages a slow drift, so $\bar{y}_t \approx m_t$ is preserved. Hence

$$\nabla L(\bar{\theta}_t) = U_k \Lambda \bar{x}_t + U_\perp M \bar{y}_t \approx U_\perp M \bar{y}_t$$

lies almost entirely in the bulk and $\chi_k \rightarrow 0$, the observed collapse. The same noise-in-sharp-directions mechanism that creates the alignment on the live iterate is what averaging removes.

b) Relation to Polyak–Ruppert averaging.: This is the geometric face of the classical optimality of iterate averaging (Polyak et al., 1992): near a minimum the average is asymptotically normal with the Newton-like covariance $H^{-1} \Sigma H^{-1} / N$, attained without ever forming H . The fluctuation energy removed is largest along the top- k (high-curvature) Hessian directions, so SWA acts like an H^{-1} preconditioner near the optimum: a reweighting of the iterate’s fluctuations, not a per-step rescaling of the update. The slightly higher training loss is consistent — SWA was switched on while the bulk drift m_t was still decaying, so the average lags the still-descending live iterate, and its benefit is expected on held-out quality (a flatter minimum) rather than on the training loss.

XIII. MOMENTUM AND THE GEOMETRY OF THE SGDM UPDATE

Song et al. (2025) (Appendix F) report the same negative result for SGD with momentum as for plain SGD: projecting the update onto the dominant Hessian subspace stops training, while the bulk projection keeps learning. This Dom/Bulk switch is binary, but momentum makes the update u_t depend on the gradient history rather than on the current gradient, so its geometry can change continuously with the momentum coefficient. As our Muon results (Fig. 6) already show, the alignment of the update need not match that of the gradient. We therefore measure, as a function of β , where the SGDM update lives relative to the dominant subspace. Throughout, β is the SGDM coefficient (not the EMA factor of the alignment score), and the alignment $\chi_k(u) = \|P_k u\|/\|u\|$ is always evaluated on the update u_t .

We train the 3-layer MLP on MNIST-5k with SGDM, learning rate $\eta = 10^{-2}$, for 10^4 steps, sweeping $\beta \in \{0.1, 0.2, \dots, 0.9\}$ (one run per value, fixed seed). The dominant subspace is the top- k Hessian eigenspace with $k = 10 = \# \text{classes}$ (Papayan, 2020), computed by full-dataset Lanczos and refreshed along the trajectory. For each run we log the update alignment $\chi_k(u_t)$ and its EMA $\bar{\chi}_k(u_t)$, the Hessian stable rank, loss/accuracy, and the number of steps to reach training loss < 0.1 .

A. The update leaves the dominant subspace as momentum grows

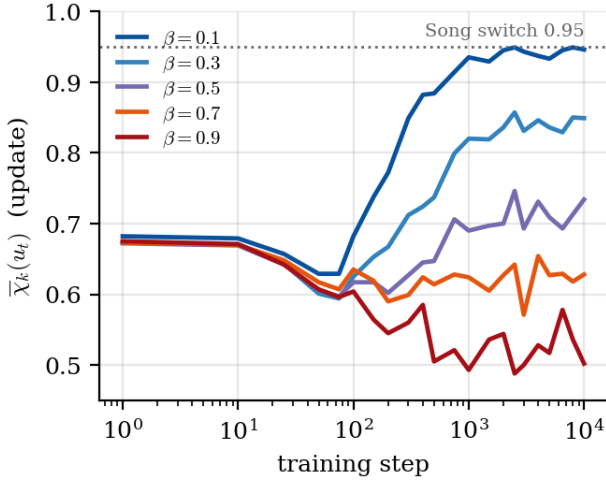


Fig. 16. As momentum grows, the SGDM update leaves the dominant subspace. EMA of the update alignment $\bar{\chi}_k(u_t)$ for SGDM on MLP/MNIST-5k. At small β the update reproduces the high gradient alignment and holds near the 0.95 switching threshold; as β grows the plateau drops, becomes flat ($\beta = 0.7$), and finally decays ($\beta = 0.9$).

The plateau of $\bar{\chi}_k(u)$ decreases monotonically across the nine runs, from 0.94 at $\beta = 0.1$ to 0.55 at $\beta = 0.9$ (Fig. 16, 17(a)): at small momentum the update reproduces the gradient level of Gur-Ari et al. (2018) and holds, while at large momentum it flattens and then decays. The classical

picture thus survives in the update only at small momentum. Optimization accelerates over the same sweep: steps to loss 0.1 fall from 3410 to 740, a $4.6\times$ speedup, with every run reaching near-perfect accuracy. The Hessian stays low-rank throughout, so the spectral gap and the $k \approx \# \text{classes}$ structure persist across the sweep.

The key contrast is gradient versus update (Fig. 17(b)). The gradient alignment $\chi_k(\nabla L)$ stays at the Gur-Ari et al. (2018) and Song et al. (2025) level, independent of β , because momentum does not touch the instantaneous gradient; only the update alignment falls. The widening gap is the fraction of the step that momentum routes out of the dominant subspace into the bulk. Since Song et al. (2025) show that the bulk component drives learning, momentum acts as an implicit bulk-amplifier, and the speedup tracks this routing (Zakarin et al., 2025). Unlike pure Bulk-SGD, which they find unstable, SGDM does not diverge, because it keeps a substantial dominant component even at the largest β . The same mechanism connects the effect to Muon (Jordan et al., 2024b), whose orthogonalization is likewise a bulk-favouring transform of the update. The decoupling here is correlational; the causal test — projecting the SGDM update onto dom versus bulk across the same β grid — is left to the projection experiments.

B. A closed-form prediction from the quadratic model

The effect extends the noise-driven quadratic analysis of Section VIII-D to the momentum buffer $u_t = \sum_{s \leq t} \beta^{t-s} g_s$, modelling the per-coordinate gradient sequence as a stationary process with lag-one autocorrelation a .

Lemma 1. *Let a per-coordinate gradient sequence be wide-sense stationary with variance σ_g^2 and autocovariance $R(\Delta) = \sigma_g^2 a^{|\Delta|}$, $a \in (-1, 1)$. For the momentum buffer $u_t = \sum_{s \leq t} \beta^{t-s} g_s$ with $0 \leq \beta < 1$, the stationary variance is*

$$\text{Var}(u_\infty) = \sigma_g^2 \frac{1 + \beta a}{(1 - \beta^2)(1 - \beta a)}.$$

In particular $\text{Var}(u_\infty) \rightarrow \sigma_g^2/(1 - \beta)^2$ as $a \rightarrow 1^-$, and $\text{Var}(u_\infty) \rightarrow \sigma_g^2/(1 + \beta)^2$ as $a \rightarrow -1^+$.

Proof. By stationarity, $\text{Var}(u_\infty) = \sum_{j, j' \geq 0} \beta^{j+j'} R(j - j') = \sigma_g^2 \sum_{j, j' \geq 0} \beta^{j+j'} a^{|j-j'|}$. Separating $j = j'$ from $j \neq j'$ and summing the geometric series gives $\sigma_g^2 \left[\frac{1}{1-\beta^2} + \frac{2}{1-\beta^2} \frac{\beta a}{1-\beta a} \right] = \sigma_g^2 \frac{1+\beta a}{(1-\beta^2)(1-\beta a)}$. The two limits follow by substitution. $\square$

The two limits give the mechanism. Bulk (flat) directions drift slowly along the valley floor, where consecutive gradients are positively correlated ($a \rightarrow 1$) and momentum amplifies their variance by $1/(1-\beta)^2$; dominant (sharp) directions oscillate across the steep walls, where gradients are anti-correlated ($a \rightarrow -1$) and momentum suppresses them by $1/(1+\beta)^2$. With $r_g = \chi_k(\nabla L)^2/(1 - \chi_k(\nabla L)^2) \approx 5.5$ for the dominant-to-bulk gradient energy ratio, and symmetric autocorrelations $a_{\text{Dom}} = -a$, $a_{\text{bulk}} = +a$, the update alignment satisfies

$$\chi_k(u)^2 = \frac{r_g q(\beta)^2}{1 + r_g q(\beta)^2}, \quad q(\beta) = \frac{1 - \beta a}{1 + \beta a}.$$

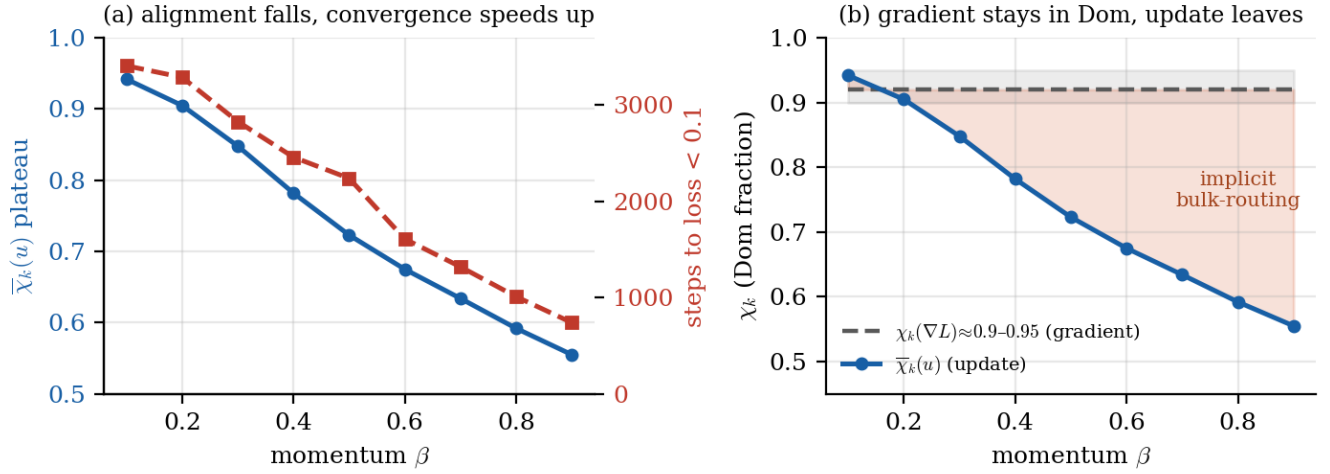


Fig. 17. **Momentum routes the step from the dominant subspace into the bulk.** Momentum sweep for SGDM on MLP/MNIST-5k. (a) As β increases, the update-alignment plateau $\bar{\chi}_k(u)$ decreases while convergence (steps to loss < 0.1) speeds up: lower alignment with the dominant subspace coincides with faster training. (b) The gradient alignment $\chi_k(\nabla L)$ stays at the Phenomenon-1 level (≈ 0.90 – 0.95 , shaded), independent of β , whereas the update alignment $\bar{\chi}_k(u)$ falls. The widening gap is the fraction of the actual step that momentum routes out of the dominant subspace into the bulk.

A single effective autocorrelation $a \approx 0.66$ reproduces the measured plateau to within ~ 0.04 over the whole sweep (Fig. 18); the idealized $a = 1$ gives the steepest possible decline and bounds the effect. A 2D simulation of Song’s $L = \frac{1}{2}(1000x^2 + y^2)$ reproduces the direction but a much milder decline, because a single sharp mode cannot stay near $a \rightarrow -1$ across the whole β range; the stronger MNIST decline requires the genuinely k -dimensional dominant subspace, over which many oscillatory directions average. The model is falsifiable: the lag-one gradient autocorrelation should be $\approx -a$ inside the dominant subspace and $\approx +a$ in the bulk, measurable directly from the projected gradient autocorrelations.

For momentum-based optimizers the informative diagnostic is the alignment of the update, not the gradient: the gradient alignment is an almost optimizer-independent artifact, while the update alignment is optimizer-dependent and tracks optimization speed. SGDM is thus a continuous, stable interpolation between SGD and Bulk-SGD, with β setting the dominant/bulk mixture of the applied step. The next steps are to co-log $\chi_k(\nabla L)$ and $\chi_k(u)$ in the same runs, and to run the Dom-/Bulk-SGDM intervention across β to turn the correlation into a causal statement.

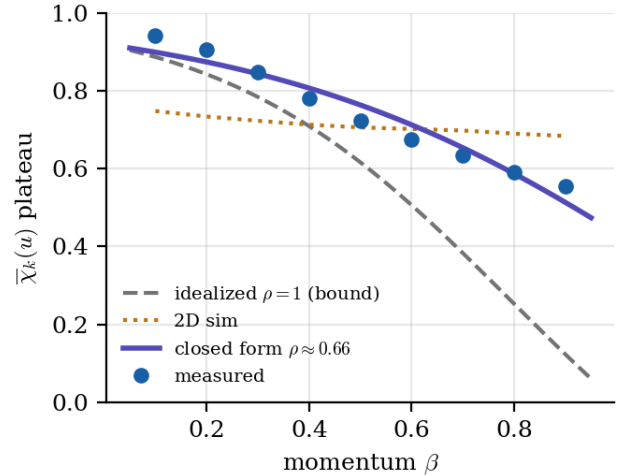


Fig. 18. **A one-parameter autocorrelation law predicts the alignment decline.** Update-alignment plateau versus momentum: measured values, the closed-form prediction $\chi_k(u)^2 = r_g q^2 / (1 + r_g q^2)$ with effective gradient autocorrelation $a \approx 0.66$, the idealized $a = 1$ bound, and a literal 2D simulation of $L = \frac{1}{2}(1000x^2 + y^2)$. The one-parameter law tracks the measured decline; the idealized limit bounds it.

REFERENCES

- Amsel, Noah, David Persson, Christopher Musco, and Robert M. Gower (2026). *The Polar Express: Optimal Matrix Sign Methods and Their Application to the Muon Algorithm*.
- Arous, Gerard Ben, Reza Gheissari, Jiaoyang Huang, and Aukosh Jagannath (2024). “High-dimensional SGD aligns with emerging outlier eigenspaces”. In: *The Twelfth International Conference on Learning Representations*.
- Cohen, Jeremy, Behrooz Ghorbani, Shankar Krishnan, Naman Agarwal, Sourabh Medapati, Michal Badura, Daniel Suo, Zachary Nado, George E. Dahl, and Justin Gilmer (2023). “Adaptive Gradient Methods at the Edge of Stability”. In: *NeurIPS 2023 Workshop Heavy Tails in Machine Learning*.
- Cohen, Jeremy, Simran Kaur, Yuanzhi Li, J Zico Kolter, and Ameet Talwalkar (2021). “Gradient Descent on Neural Networks Typically Occurs at the Edge of Stability”. In: *International Conference on Learning Representations*.
- Damian, Alex, Eshaan Nichani, and Jason D. Lee (2023). “Self-Stabilization: The Implicit Bias of Gradient Descent at the Edge of Stability”. In: *International Conference on Learning Representations (ICLR)*.
- Gauch, Martin, Maximilian Beck, Thomas Adler, Dmytro Koturs, Stefan Fiel, Hamid Eghbal-zadeh, Johannes Brandstetter, Johannes Kofler, Markus Holzleitner, Werner Zellinger, Daniel Klotz, Sepp Hochreiter, and Sebastian Lehner (2022). “Few-Shot Learning by Dimensionality Reduction in Gradient Space”. In: *Proceedings of The 1st Conference on Lifelong Learning Agents*.
- Ghorbani, Behrooz, Shankar Krishnan, and Ying Xiao (2019). “An Investigation into Neural Net Optimization via Hessian Eigenvalue Density”. In: *Proceedings of the 36th International Conference on Machine Learning*.
- Grishina, Ekaterina, Matvey Smirnov, and Maxim Rakhuba (2026). *Accelerating Newton-Schulz Iteration for Orthogonalization via Chebyshev-type Polynomials*.
- Gur-Ari, Guy, Daniel A. Roberts, and Ethan Dyer (2018). *Gradient Descent Happens in a Tiny Subspace*.
- Horn, Roger A. and Charles R. Johnson (2013). *Matrix Analysis*.
- Izmailov, Pavel, Dmitrii Podoprikin, Timur Garipov, Dmitry Vetrov, and Andrew Gordon Wilson (2018). “Averaging Weights Leads to Wider Optima and Better Generalization”. In: *Conference on Uncertainty in Artificial Intelligence (UAI)*.
- Jordan, Keller, Yuchen Jin, Vlado Boza, Jiacheng You, Franz Cesista, Laker Newhouse, and Jeremy Bernstein (2024a). *Muon: An optimizer for hidden layers in neural networks*.
- (2024b). *Muon: An optimizer for hidden layers in neural networks*.
- Kingma, Diederik P. and Jimmy Ba (2015). “Adam: A Method for Stochastic Optimization”. In: *International Conference on Learning Representations (ICLR)*.
- Kostenetskiy, P. S., R. A. Chulkevich, and V. I. Kozyrev (2021). “HPC Resources of the Higher School of Economics”. In: *Journal of Physics: Conference Series* 1740, p. 012050.
- Krizhevsky, Alex (2009). *Learning multiple layers of features from tiny images*.
- Kunstner, Frederik, Philipp Hennig, and Lukas Balles (2019). “Limitations of the Empirical Fisher Approximation for Natural Gradient Descent”. In: *Advances in Neural Information Processing Systems (NeurIPS)*.
- LeCun, Yann (1998). “The MNIST database of handwritten digits”. In: <http://yann.lecun.com/exdb/mnist/>.
- Li, Chunyuan, Heerad Farkhoor, Rosanne Liu, and Jason Yosinski (2018). *Measuring the Intrinsic Dimension of Objective Landscapes*.
- Li, Tao, Lei Tan, Qinghua Tao, Yipeng Liu, and Xiaolin Huang (2021). *Low Dimensional Landscape Hypothesis is True: DNNs can be Trained in Tiny Subspaces*.
- Mahdavinia, Pouria and Mehrdad Mahdavi (2025). *Low-rank Momentum Factorization for Memory Efficient Training*.
- Papayan, Vardan (2019). “Measurements of Three-Level Hierarchical Structure in the Outliers in the Spectrum of Deepnet Hessians”. In: *CoRR*.
- (2020). “Traces of Class/Cross-Class Structure Pervade Deep Learning Spectra”. In: *Journal of Machine Learning Research* 21.252.
- Polyak, Boris T. and Anatoli B. Juditsky (1992). “Acceleration of Stochastic Approximation by Averaging”. In: *SIAM Journal on Control and Optimization* 30.4.
- Sagun, Levent, Leon Bottou, and Yann LeCun (2017). *Eigenvalues of the Hessian in Deep Learning: Singularity and Beyond*.
- Sagun, Levent, Utku Evci, V. Ugur Guney, Yann Dauphin, and Leon Bottou (2018). *Empirical Analysis of the Hessian of Over-Parametrized Neural Networks*.
- Singh, Sidak Pal, Gregor Bachmann, and Thomas Hofmann (2021). “Analytic Insights into Structure and Rank of Neural Network Hessian Maps”. In: *Advances in Neural Information Processing Systems*. Ed. by A. Beygelzimer, Y. Dauphin, P. Liang, and J. Wortman Vaughan.
- Socher, Richard, Alex Perelygin, Jean Wu, Jason Chuang, Christopher D. Manning, Andrew Ng, and Christopher Potts (2013). “Recursive Deep Models for Semantic Compositionality Over a Sentiment Treebank”. In: *Proceedings of the 2013 Conference on Empirical Methods in Natural Language Processing*.
- Song, Minhak, Kwangjun Ahn, and Chulhee Yun (2025). “Does SGD really happen in tiny subspaces?” In: *International Conference on Learning Representations*.
- Zakarin, Daniyar and Sidak Pal Singh (2025). *Accelerating Neural Network Training Along Sharp and Flat Directions*.
- Zhang, Zhong, Bang Liu, and Junming Shao (2023). “Fine-tuning Happens in Tiny Subspaces: Exploring Intrinsic Task-specific Subspaces of Pre-trained Language Models”. In: *Proceedings of the 61st Annual Meeting of the As*

sociation for Computational Linguistics (Volume 1: Long Papers).

Zhao, Jiawei, Zhenyu Zhang, Beidi Chen, Zhangyang Wang, Anima Anandkumar, and Yuandong Tian (2024). “Ga-Lore: Memory-Efficient LLM Training by Gradient Low-Rank Projection”. In: *International Conference on Machine Learning*.

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APPENDIX A GITHUB REPOSITORY

The source code for the project is available at:
[GitHub repository](#)

APPENDIX B PROOF OF THEOREM 1

Proof. Let $U_k \in \mathbb{R}^{d \times k}$ denote the matrix of an orthonormal basis of the dominant subspace, and let $U_\perp \in \mathbb{R}^{d \times (d-k)}$ be the basis of the orthogonal complement.

Decompose the parameters relative to the minimum point θ^*

$$\theta - \theta^* = U_k x + U_\perp y, \quad x \in \mathbb{R}^k, \quad y \in \mathbb{R}^{d-k}$$

Here x are the “dominant / sharp” coordinates, and y are the “bulk / flat” coordinates.

A. Simple Quadratic model

Consider

$$L(\theta) = \frac{1}{2} x^\top \Lambda x + \frac{1}{2} y^\top M y,$$

where

$$\Lambda = \text{diag}(\lambda_1, \dots, \lambda_k) \quad M = \text{diag}(\mu_{k+1}, \dots, \mu_d),$$

and

$$\lambda_1 \geq \dots \geq \lambda_k \gg \mu_{k+1} \geq \dots \geq \mu_d > 0$$

Equivalently,

$$L(\theta) = \frac{1}{2} (\theta - \theta^*)^\top H (\theta - \theta^*), \quad H = U_k \Lambda U_k^\top + U_\perp M U_\perp^\top$$

Then

$$\nabla L(\theta) = H(\theta - \theta^*) = U_k \Lambda x + U_\perp M y$$

$$\nabla^2 L(\theta) = H$$

To make the notion of approximate low-rankness precise, consider the best rank- k spectral truncation of the Hessian,

$$H_k := U_k \Lambda U_k^\top \Rightarrow H - H_k = U_\perp M U_\perp^\top$$

Hence the Hessian is approximately rank- k in the spectral sense whenever

$$\frac{\|H - H_k\|_2}{\|H\|_2} = \frac{\mu_{k+1}}{\lambda_1} \ll 1$$

Equivalently, after removing the leading k eigendirections, the remaining curvature is uniformly small in every direction.

B. Gradient evolution in continuous and discrete time

First, consider the continuous gradient flow:

$$\dot{\theta}(t) = -\nabla L(\theta(t))$$

In the (x, y) coordinates, this is simply

$$\dot{x}(t) = -\Lambda x(t), \quad \dot{y}(t) = -M y(t).$$

Therefore,

$$x_i(t) = x_i(0) e^{-\lambda_i t}, \quad y_j(t) = y_j(0) e^{-\mu_j t}$$

The components of the full gradient are

$$g_i(t) = \lambda_i x_i(0) e^{-\lambda_i t}, \quad g_j(t) = \mu_j y_j(0) e^{-\mu_j t}.$$

Lemma 2. Suppose $\lambda_k > \mu_{k+1}$ and the initial point has a nonzero bulk component, i.e. $\exists j > k : y_j(0) \neq 0$. Then in this model noiseless gradient flow does not align the full gradient with the top- k subspace. On the contrary, at late times,

$$\chi_k(\nabla L(\theta(t))) \rightarrow 0$$

Proof. For any sharp component $i \leq k$ and bulk component $j > k$,

$$\frac{|g_i(t)|}{|g_j(t)|} = \frac{|\lambda_i x_i(0)|}{|\mu_j y_j(0)|} e^{-(\lambda_i - \mu_j)t}$$

Since $\lambda_i - \mu_j > 0$, every such pairwise ratio decays exponentially. To turn this into a statement about χ_k itself, write the squared alignment as a ratio of sharp and bulk gradient energies, $\chi_k^2(\nabla L(\theta(t))) = S(t)/(S(t) + B(t))$, where

$$S(t) = \sum_{i=1}^k \lambda_i^2 x_i(0)^2 e^{-2\lambda_i t}, \quad B(t) = \sum_{j=k+1}^d \mu_j^2 y_j(0)^2 e^{-2\mu_j t}.$$

Let $\lambda_{\min} = \min\{\lambda_i : x_i(0) \neq 0\} \geq \lambda_k$ and $\mu_{\min}^+ = \min\{\mu_j : y_j(0) \neq 0\}$, the latter well defined by hypothesis. Then $S(t) = O(e^{-2\lambda_{\min} t})$ and $B(t) = \Theta(e^{-2\mu_{\min}^+ t})$, hence $S(t)/B(t) = O(e^{-2(\lambda_{\min} - \mu_{\min}^+)t})$. Since $\lambda_{\min} \geq \lambda_k > \mu_{k+1} \geq \mu_{\min}^+$, the exponent is strictly negative, so $S/B \rightarrow 0$ and therefore $\chi_k^2 = S/(S + B) \rightarrow 0$. Equivalently, each sharp gradient component decays strictly faster than the slowest surviving bulk component, so the bulk dominates asymptotically. $\square$

□

Thus, deterministic gradient flow on a fixed quadratic objective does not by itself explain top-k gradient alignment. Since sharp components decay faster than bulk components, the long-time gradient is instead dominated by the low-curvature directions. Therefore, any observed concentration of the full gradient in the dominant eigenspace under small-step SGD requires an additional mechanism. In our model, this mechanism is minibatch-induced stochasticity, whose perturbations are amplified by the sharp curvature directions.

Now consider discrete SGD. Suppose the update uses an unbiased stochastic gradient:

$$\theta_{t+1} = \theta_t - \eta(\nabla L(\theta_t) + \xi_t), \quad \mathbb{E}[\xi_t | \theta_t] = 0$$

Here ξ_t is minibatch noise term. Importantly, alignment is measured for the full gradient $\nabla L(\theta_t)$, while the noise acts through the trajectory θ_t itself, constantly perturbing it away from the bottom of the valley. In coordinates:

$$x_{t+1} = (I - \eta\Lambda)x_t - \eta\xi_t^{(x)}, \quad y_{t+1} = (I - \eta M)y_t - \eta\xi_t^{(y)}$$

Assume that the noise coordinates are independent and

$$\text{Var}(\xi_{t,i}^{(x)}) = \sigma_i^2, \quad \text{Var}(\xi_{t,j}^{(y)}) = \zeta_j^2$$

Lemma 3. *For any coordinate with curvature $\lambda > 0$ and*

$$u_{t+1} = (1 - \eta\lambda)u_t - \eta\tau_t, \quad \mathbb{E}\tau_t = 0, \quad \text{Var}(\tau_t) = s^2,$$

if $0 < \eta\lambda < 2$, then the stationary variance is

$$\text{Var}(u_\infty) = \frac{\eta s^2}{2\lambda - \eta\lambda^2}$$

and stationary energy of the corresponding full-gradient component λu_t is

$$\mathbb{E}[(\lambda u_\infty)^2] = \frac{\eta\lambda s^2}{2 - \eta\lambda}$$

Therefore, for a fixed level of update noise, the gradient energy generated by the noise increases with the curvature λ .

Proof. Let $v = \text{Var}(u_\infty)$. For $0 < \eta\lambda < 2$ the mean decays as $\mathbb{E}[u_t] = (1 - \eta\lambda)^t \mathbb{E}[u_0] \rightarrow 0$, so the stationary energy coincides with the stationary variance. Since the noise τ_t is independent of the current iterate u_t , taking variances in the recursion yields $\text{Var}(u_{t+1}) = (1 - \eta\lambda)^2 \text{Var}(u_t) + \eta^2 s^2$, a contraction whose stationary fixed point v satisfies

$$v = (1 - \eta\lambda)^2 v + \eta^2 s^2$$

Hence,

$$v = \frac{\eta^2 s^2}{1 - (1 - \eta\lambda)^2} = \frac{\eta^2 s^2}{2\eta\lambda - \eta^2\lambda^2} = \frac{\eta s^2}{2\lambda - \eta\lambda^2}$$

Multiplying by λ^2 , we obtain

$$\mathbb{E}[(\lambda u_\infty)^2] = \lambda^2 v = \frac{\eta\lambda s^2}{2 - \eta\lambda}$$

A small noisy displacement in a sharp direction has scale

$$x_i^{\text{noise}} = \mathcal{O}\left(\sqrt{\frac{\eta}{\lambda_i}}\right)$$

but after multiplication by the Hessian, the full gradient receives a sharp component of scale

$$g_i^{\text{noise}} = \lambda_i x_i^{\text{noise}} = \mathcal{O}\left(\sqrt{\eta\lambda_i}\right)$$

Thus, in parameter space the displacement is small, but in gradient space it is significant precisely because the curvature is large. For bulk directions, similarly,

$$g_j^{\text{noise}} = \mathcal{O}\left(\sqrt{\eta\mu_j}\right)$$

which is much smaller when $\lambda_i \gg \mu_j$.

Introduce an energy-based surrogate for alignment, defined as a ratio of expected energies. Because it uses a ratio of expectations rather than the expectation of a ratio, it is a tractable surrogate for $\mathbb{E}[\chi_k^2(\nabla L(\theta_t))]$; note that it is quadratic in the alignment χ_k .

$$\bar{\chi}_k(t) := \frac{\mathbb{E}\|P_k \nabla L(\theta_t)\|^2}{\mathbb{E}\|\nabla L(\theta_t)\|^2}$$

In our model, expanding in the (x, y) coordinates gives the exact expression

$$\bar{\chi}_k(t) = \frac{\sum_{i=1}^k \lambda_i^2 \mathbb{E}[x_{t,i}^2]}{\sum_{i=1}^k \lambda_i^2 \mathbb{E}[x_{t,i}^2] + \sum_{j=k+1}^d \mu_j^2 \mathbb{E}[y_{t,j}^2]}$$

After the relaxation of the sharp coordinates, we obtain the approximation

$$\bar{\chi}_k(t) \approx \frac{\sum_{i=1}^k \frac{\eta\lambda_i \sigma_i^2}{2 - \eta\lambda_i}}{\sum_{i=1}^k \frac{\eta\lambda_i \sigma_i^2}{2 - \eta\lambda_i} + \sum_{j=k+1}^d \left(\mu_j^2 m_{t,j}^2 + \frac{\eta\mu_j \zeta_j^2}{2 - \eta\mu_j} \right)}$$

where

$$m_t = \mathbb{E}[y_t]$$

This shows that if spectral gap is large, even a small amount of noise can make the observed full gradient almost entirely dominant, even though the mean drift in the bulk has not yet fully decayed.

In the symmetric case

$$\Lambda = \lambda I_k, \quad M = \mu I_{d-k}$$

with noise

$$\sigma_i = \zeta_j = s$$

we obtain a particularly simple formula for the purely noise-driven regime:

$$\bar{\chi}_k^{\text{noise}} = \frac{k\lambda/(2-\eta\lambda)}{k\lambda/(2-\eta\lambda) + (d-k)\mu/(2-\eta\mu)} \approx \frac{k\lambda}{k\lambda + (d-k)\mu}.$$

Therefore, when

$$\kappa = \frac{\lambda}{\mu} \gg \frac{d-k}{k},$$

the gradient energy can be concentrated almost entirely in the top-k subspace, although this may largely be a noise-induced response of the trajectory to sharp curvature.

C. Why projection onto the dominant Hessian subspace can stop training

For projected gradient flow

$$\dot{\theta}(t) = -P_k \nabla L(\theta(t))$$

always holds

$$\frac{d}{dt}L(\theta(t)) = -\nabla L(\theta(t))^\top P_k \nabla L(\theta(t)) = -\|P_k \nabla L(\theta(t))\|^2$$

That is, the projected flow decreases the loss only through the dominant component of the full gradient.

Theorem 2. *For the quadratic model*

$$L(\theta) = \frac{1}{2}x^\top \Lambda x + \frac{1}{2}y^\top M y$$

projected gradient flow on the dominant subspace

$$\dot{\theta} = -P_k \nabla L(\theta)$$

and projected SGD

$$\theta_{t+1} = \theta_t - \eta P_k (\nabla L(\theta_t) + \xi_t)$$

freeze the bulk coordinates y. In particular:

$$\dot{x} = -\Lambda x, \quad \dot{y} = 0$$

for continuous time, and

$$x_{t+1} = (I - \eta \Lambda)x_t - \eta \xi_t^{(x)}, \quad y_{t+1} = y_t$$

for discrete time.

Therefore:

1) *The set of stationary points of the projected flow is*

$$\mathcal{M}_{\text{proj}} = \{\theta : P_k \nabla L(\theta) = 0\} = \{\theta^* + U_\perp y : y \in \mathbb{R}^{d-k}\}$$

that is, an entire $(d-k)$ -dimensional manifold.

2) *Every such point with $y \neq 0$ is suboptimal for the original loss, because*

$$\nabla L(\theta^* + U_\perp y) = U_\perp M y \neq 0 \quad \text{for } y \neq 0$$

3) *If the switch to projected dynamics happens at time t_0 , then for projected flow*

$$L(t) \rightarrow L_\infty = \frac{1}{2}y(t_0)^\top M y(t_0) > 0 \quad \text{for } y(t_0) \neq 0$$

4) *For projected SGD with independent noise of variances σ_i^2 in the dominant coordinates and $0 < \eta < 2/\lambda_1$*

$$\lim_{t \rightarrow \infty} \mathbb{E}L(\theta_t) = \frac{1}{2}y_0^\top M y_0 + \frac{1}{2} \sum_{i=1}^k \frac{\eta \sigma_i^2}{2 - \eta \lambda_i}$$

That is, a positive noise floor appears on top of the bulk-loss floor.

Proof. Since $P_k U_k = U_k$ and $P_k U_\perp = 0$, we have

$$P_k \nabla L(\theta) = U_k U_k^\top (U_k \Lambda x + U_\perp M y) = U_k \Lambda x$$

Therefore,

$$\dot{\theta} = -U_k \Lambda x$$

Multiplying on the left by $U_k^\top$ and $U_\perp^\top$, we obtain

$$\dot{x} = -\Lambda x, \quad \dot{y} = 0$$

Hence,

$$x(t) = x(t_0)e^{-\Lambda(t-t_0)}, \quad y(t) \equiv y(t_0)$$

and therefore

$$\begin{aligned} L(t) &= \frac{1}{2}x(t_0)^\top e^{-2\Lambda(t-t_0)} \Lambda x(t_0) + \frac{1}{2}y(t_0)^\top M y(t_0) \\ &\rightarrow \frac{1}{2}y(t_0)^\top M y(t_0). \end{aligned}$$

For projected SGD, the calculation is the same at the coordinate level:

$$x_{t+1} = (I - \eta \Lambda)x_t - \eta \xi_t^{(x)}, \quad y_{t+1} = y_t$$

Thus the bulk is indeed frozen, $y_t \equiv y_0$. For each dominant coordinate the mean decays to zero, so $\mathbb{E}[x_{t,i}^2] \rightarrow \text{Var}(x_{\infty,i})$, and the stationary variance follows from the previous lemma (applicable since $0 < \eta \lambda_i < 2$ for all i , which is exactly the assumption $0 < \eta < 2/\lambda_1$), after which

$$\mathbb{E} \left[\frac{1}{2} \lambda_i x_{t,i}^2 \right] \rightarrow \frac{1}{2} \lambda_i \cdot \frac{\eta \sigma_i^2}{2 \lambda_i - \eta \lambda_i^2} = \frac{1}{2} \frac{\eta \sigma_i^2}{2 - \eta \lambda_i}.$$

Summing over i , we obtain the claimed limit for the expected loss. $\square$

This theorem gives a rigorous mechanism within this model to the question “why can training stop under projection?” The reason is not that projected dynamics has no local loss decrease at all, but that after the fast sharp transient it solves a different stationary problem:

$$P_k \nabla L(\theta) = 0, \quad \text{instead of} \quad \nabla L(\theta) = 0$$

This is why alignment of the full gradient with top-k does not guarantee optimality of the projected method.

The a one-step Taylor argument looks at the current $\chi_k(\nabla L)$, but ignores that projected dynamics changes the future generation of the trajectory: the system

$$(\dot{x}, \dot{y}) = (-\Lambda x, -My)$$

turns into

$$(-\Lambda x, 0)$$

So the question is not only how large the current dominant component of the gradient is, but which coordinates continue to evolve at all after projection.

For completeness, it is useful to note the dual picture. If we project not onto the dominant subspace, but onto the bulk, then we get

$$\dot{x} = 0, \quad \dot{y} = -My$$

or in discrete form,

$$x_{t+1} = x_t, \quad y_{t+1} = (I - \eta M)y_t - \eta \xi_t^{(y)}$$

If the switch happens after the sharp coordinates are already small, then freezing x costs almost nothing, while the optimization in the bulk coordinates is preserved. Therefore Bulk-SGD can continue to work almost like the original SGD — exactly what was empirically shown in the Song et al. (2025)

APPENDIX C

METRIC-INVARIANCE OF THE DOM/BULK CONSTRUCTION

All three Hessian-subspace projectors of Section VI share the whitened Hessian $\tilde{H} = \mathcal{M}^{-1/2} H \mathcal{M}^{-1/2}$ for an SPD metric $\mathcal{M} \succ 0$. With the Gauss-Newton split $H = H_{\text{GN}} + H_F$, $H_{\text{GN}} = \frac{1}{N} \sum_{i=1}^N J_i^\top (\nabla_z^2 \ell_i) J_i$ — J_i the per-example output Jacobian, $\nabla_z^2 \ell_i$ the logit-space loss Hessian, N the number of training examples, k the number of classes — the following three statements isolate which properties of the construction survive the change of metric.

Lemma 4 (Preconditioning is a congruence). *For any metric $\mathcal{M} \succ 0$: (i) $\tilde{H}$ is congruent to H and similar to $\mathcal{M}^{-1}H$. Its eigenvectors solve the generalized eigenproblem $Hv = \lambda \mathcal{M}v$ and are $\mathcal{M}$ -orthonormal, so the dom/bulk split is $\mathcal{M}$ -orthogonal. (ii) Whitening preserves the Gauss-Newton split,*

$$\tilde{H} = \tilde{H}_{\text{GN}} + \tilde{H}_F, \quad \tilde{H}_{\text{GN}} = \frac{1}{N} \sum_{i=1}^N \tilde{J}_i^\top (\nabla_z^2 \ell_i) \tilde{J}_i,$$

acting only on the Jacobian, $\tilde{J}_i = J_i \mathcal{M}^{-1/2}$, while the output-space factor $\nabla_z^2 \ell_i$ is unchanged.

Proof. $\mathcal{M}^{-1/2}$ is symmetric and invertible, so $A \mapsto \mathcal{M}^{-1/2} A \mathcal{M}^{-1/2}$ is a congruence; equivalently $\tilde{H}$ is the Hessian in the coordinates $y = \mathcal{M}^{1/2} \theta$, under which the network as a map of the data is unchanged. Similarity to $\mathcal{M}^{-1}H$ follows from $\mathcal{M}^{1/2}(\mathcal{M}^{-1}H)\mathcal{M}^{-1/2} = \tilde{H}$; if $\tilde{H}w = \lambda w$ then $v = \mathcal{M}^{-1/2}w$ solves $Hv = \lambda \mathcal{M}v$ with $v_i^\top \mathcal{M}v_j = w_i^\top w_j$, giving the $\mathcal{M}$ -orthonormality. Substituting H_{GN} and using $\mathcal{M}^{-1/2} J_i^\top = (J_i \mathcal{M}^{-1/2})^\top$ gives (ii). $\square$

Theorem 3 (Metric-invariance of Hessian rank). *For every $\mathcal{M} \succ 0$, whitening preserves the rank structure of the Hessian: (a) $\tilde{H}$ has the same rank and inertia as H (Sylvester’s law of inertia (Horn et al., 2013)); (b) $\text{rank}(\tilde{H}_{\text{GN}}) = \text{rank}(H_{\text{GN}}) \leq N(k-1)$ for softmax cross-entropy ($\leq Nk$ for the square loss); (c) for deep linear networks the outer-product, functional and total Hessian-rank formulas of Singh et al. (2021) are unchanged.*

Proof. (a) Congruent symmetric matrices share their inertia, hence their rank (Lemma 4).

(b) By Lemma 4, $\tilde{H}_{\text{GN}} = \frac{1}{N} \sum_i \tilde{J}_i^\top S_i \tilde{J}_i$ with $S_i = \nabla_z^2 \ell_i$, $\text{rank}(S_i) \leq k-1$ for cross-entropy (null space $\text{span}\{\mathbf{1}_k\}$); since $\mathcal{M}^{-1/2}$ is invertible, $\text{rank}(\tilde{J}_i^\top S_i \tilde{J}_i) = \text{rank}(J_i^\top S_i J_i)$, and subadditivity of rank gives $\text{rank}(\tilde{H}_{\text{GN}}) \leq \sum_i \text{rank}(S_i) \leq N(k-1)$. (c) The ranks of H_o , H_f and $\dim(\text{col } H_o \cap \text{col } H_f)$ are preserved by the invertible map $\mathcal{M}^{-1/2}$. $\square$

Theorem 4 (Metric-invariance of the class-component dimension). *Let $\mathcal{C} = \text{span}\{\mu_1, \dots, \mu_k\}$ be the class-means subspace of Pappayan (2019), spanned by the class-conditional means $\mu_c = \frac{1}{|I_c|} \sum_{i \in I_c} g_i$ of the per-example gradients $g_i = \nabla_\theta \ell_i$, with $\dim \mathcal{C} \leq k$. For any $\mathcal{M} \succ 0$, whitening maps it to $\tilde{\mathcal{C}} = \mathcal{M}^{-1/2} \mathcal{C}$, with $\dim \tilde{\mathcal{C}} = \dim \mathcal{C} \leq k$. Hence the candidate dominant dimension $k = \#\text{classes}$ is metric-invariant.*

Proof. Whitening is linear, $\tilde{g}_i = \mathcal{M}^{-1/2} g_i$, so the class means transform as $\tilde{\mu}_c = \mathcal{M}^{-1/2} \mu_c$; an invertible linear map preserves the dimension of a span. $\square$